



A+ LAB SERIES V5

Lab 14: Disk Maintenance and Data Recovery

Document Version: **2026-01-02**

Material in this Lab Aligns to the Following	
CompTIA A+ (220-1201) Exam Objectives	
CompTIA A+ (220-1202) Exam Objectives	1.4: Given a scenario, use Microsoft Windows operating system features and tools. 4.2: Given a scenario, apply change management procedures. 4.3: Given a scenario, implement workstation backup and recovery methods.

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Introduction

The *Windows* operating system sometimes behaves like the reality TV show “*Hoarders*” ... it never throws anything away. Temporary files, browser cache, cookies, and other bits and pieces are left on the disk. While these files “typically” do not represent a security or operational problem, they do take up space on the hard drives. So, periodically, we need to do a thorough cleaning.

On a different tack, another aspect of being a desktop support technician is protecting the system from various accidental (user and/or application) issues. A highly skilled tech should know how to set up and use the Windows tools that can create a safe refuge where the Operating System (OS) can be restored in the event one of these problems arises.

While on the subject of stuff stored on our disks, one of the cardinal rules of desktop technical support is: backup.

Computers, in general, and hard drives, in particular, can and will fail, and if you are an adherent to *Murphy's Law*, then they will fail at the most inconvenient time. Since the media cannot be separated from the drive itself, if the drive fails, the data is lost (though there are very expensive methods to attempt to recover it from a failed drive). The best recourse, of course, is to back up the data regularly.

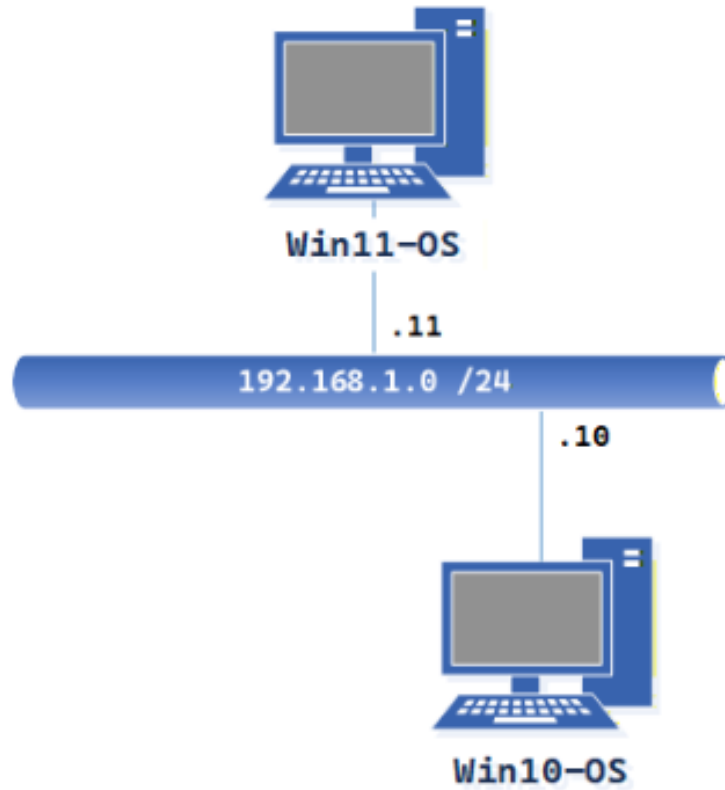
Objective

The objective of this lab is to perform the cleanup and maintenance operations required by the OS to keep it running at peak efficiency, to be prepared in the event of system-level problems, and to review key concepts related to disk maintenance and data recovery.

In this lab, you will be performing the following tasks:

- Using the Windows *Disk Cleanup* utility to clean out unneeded files
- Using Task Scheduler to schedule Disk Cleanup
- Using Windows *Disk Defragmentation* to optimize the space on the partitions
- Using Windows *Restore Points* to create states that the system can be returned to by changing the Desktop Appearance and adding files to the Desktop
- Using Windows *Backup and Restore* utilities to back up the computer to a previous state
- Backup Windows 10 files using File History to Windows 11
- Locate the Windows 10 backup on Windows 11

Lab Topology



Lab Settings

The information in the table below will be needed in order to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account (if needed)	Password (if needed)
Win10-OS	192.168.1.10 / 24	sysadmin	NDGLabpass123!
Win11-OS	192.168.1.11 / 24	sysadmin	NDGLabpass123!

1 Running the Disk Cleanup Utility

Many programs add temporary files that are no longer needed. Web browsers also add cache and cookies that can accumulate on a system. Using tools to remove these files is a good practice.

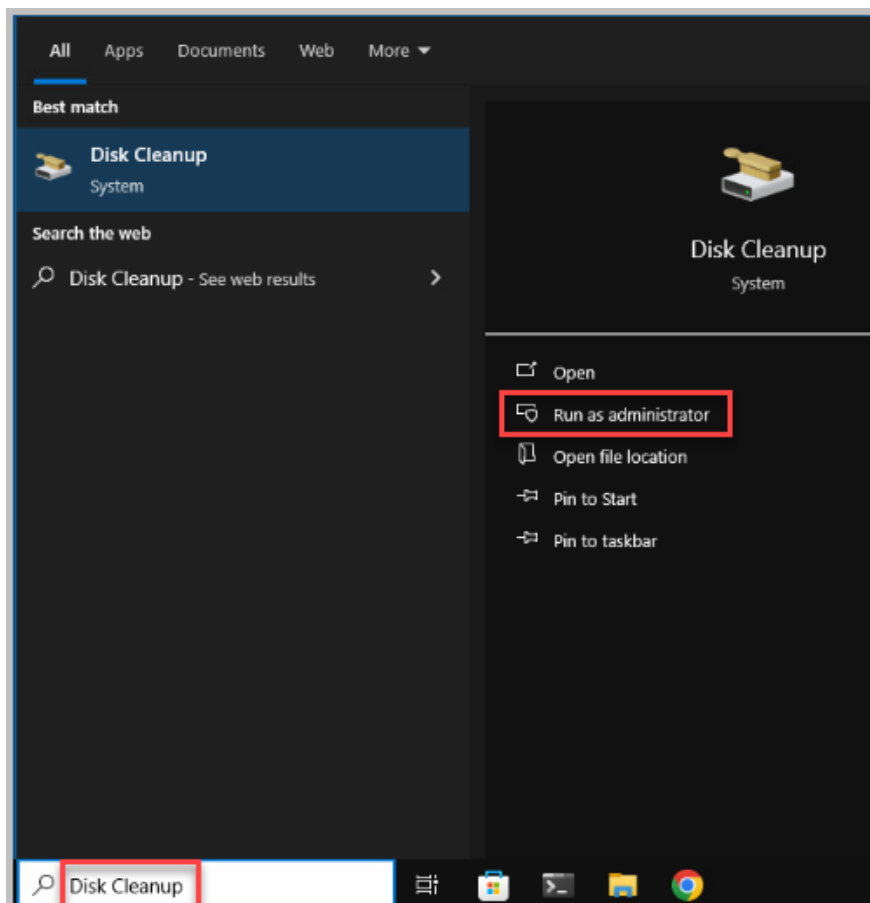
1.1 Running Disk Cleanup on Win10-OS

1. Open a console connection to the **Win10-OS** System.

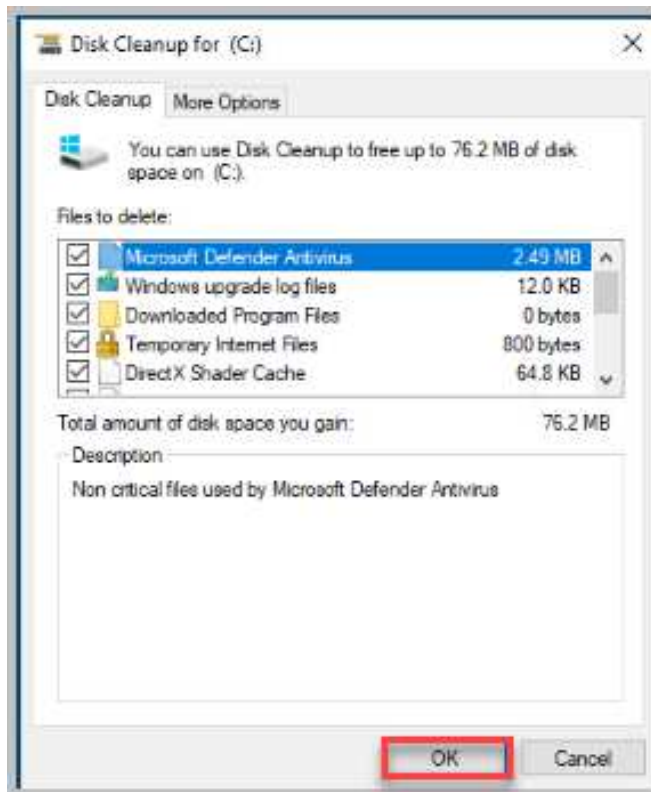


To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device's corresponding image.

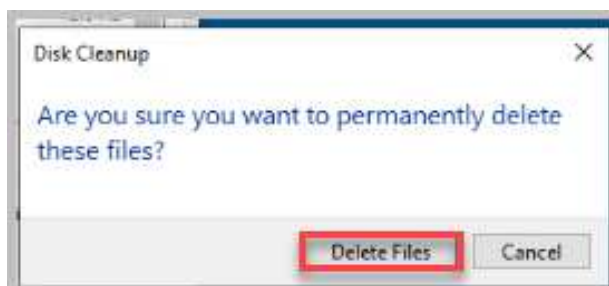
2. Type **Disk Cleanup** in the search box of the taskbar. Select **Run as administrator**.



3. Notice that the *Disk Cleanup* utility appears and starts scanning the system. Wait until the utility finishes scanning.
4. Once the *Disk Cleanup* utility finishes scanning, notice the checkboxes in the *Files to delete* pane. These indicate which areas have files that can be cleaned up and how much disk space can be recovered. Under normal operations, the users can be taught to perform basic disk cleaning periodically. Leave the defaulted checkboxes and review each item to understand what is being cleaned. Clicking on each entry will show a description. Notice the total amount of disk space to be cleaned. Click **OK** to initiate the cleanup process.



5. When prompted, click **Delete Files** to confirm the process.



- The process should take about 2-3 minutes to complete. Once completed, the utility will close automatically.



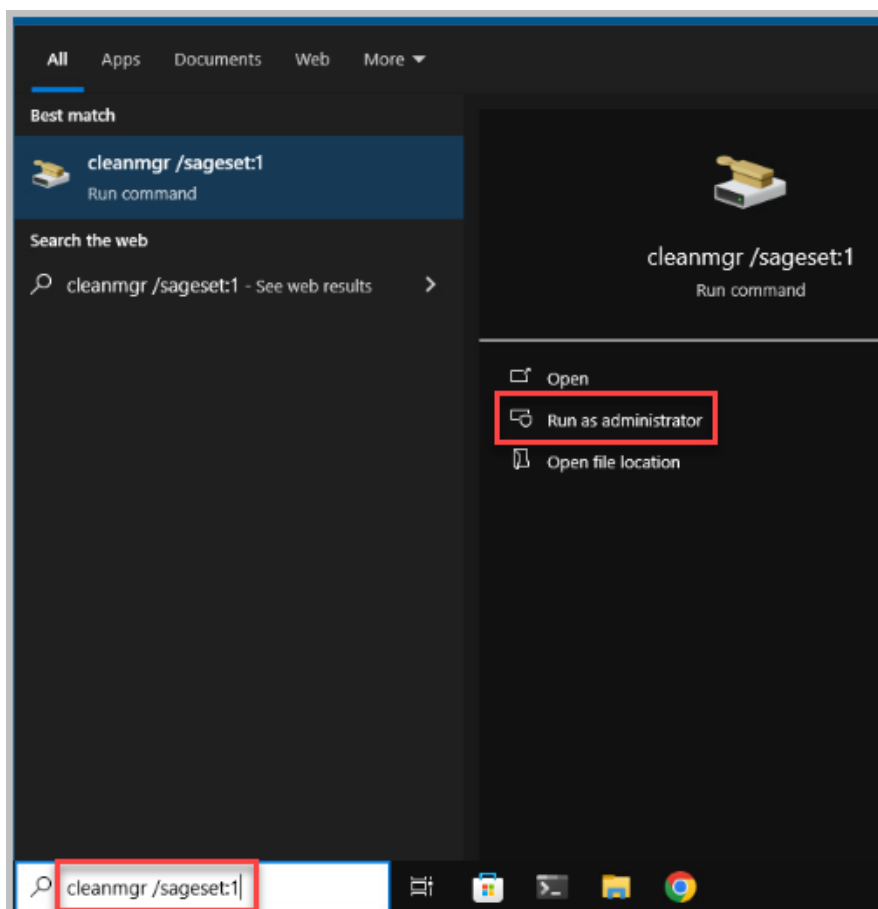
Challenge Task:

As a challenge, try to recreate this task in a different version of Windows OS that is made available in this lab environment. This will help show similarities and differences between the versions.

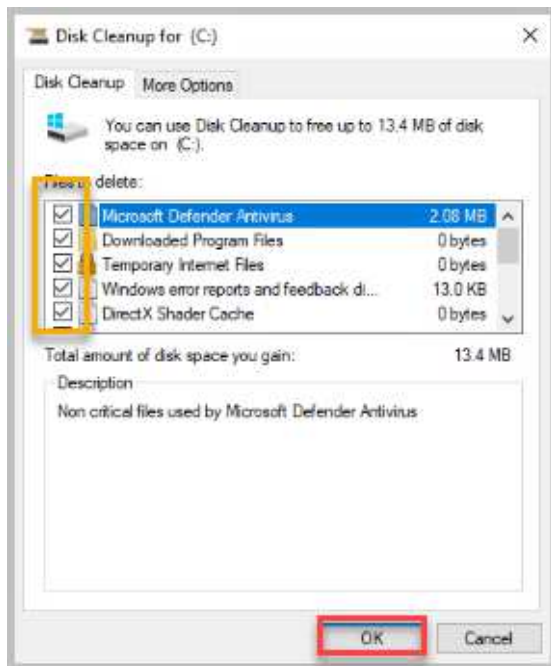
- Leave the *Win10-OS* window open to continue with the next task.

1.2 Scheduling Disk Cleanup for Win10-OS

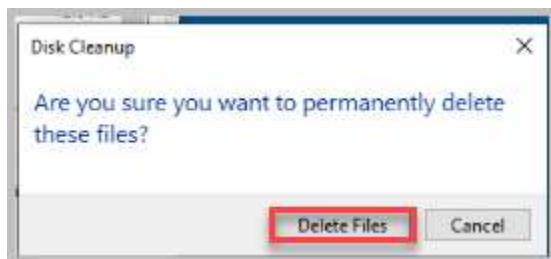
- Type **cleanmgr /sageset:1** in the search box of the taskbar. Select **Run as administrator**.



2. Once the *Disk Cleanup Settings* window appears, check all of the checkboxes and click **OK**.

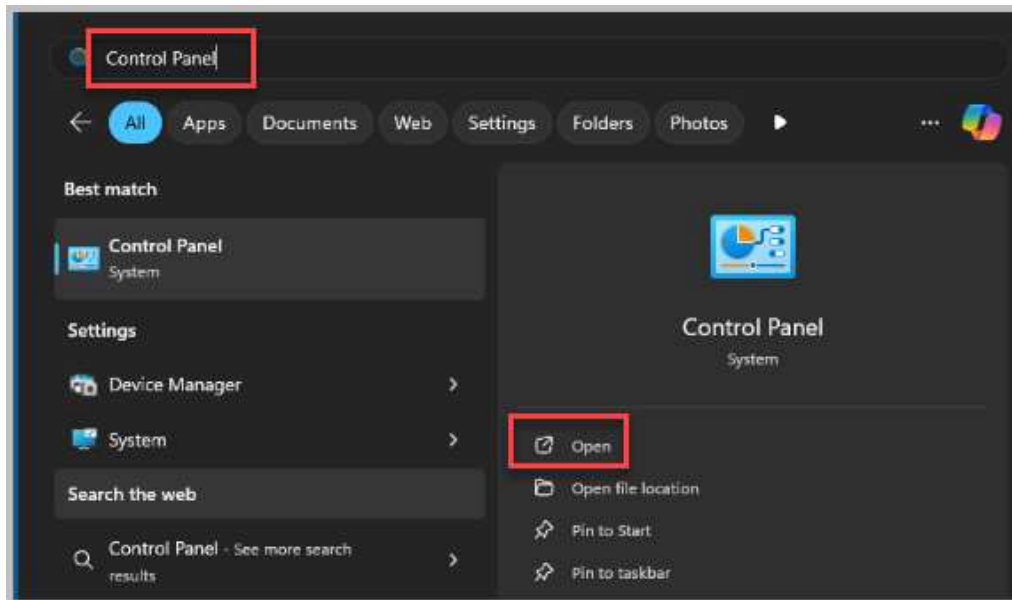


3. When prompted, click **Delete Files** to confirm the process.

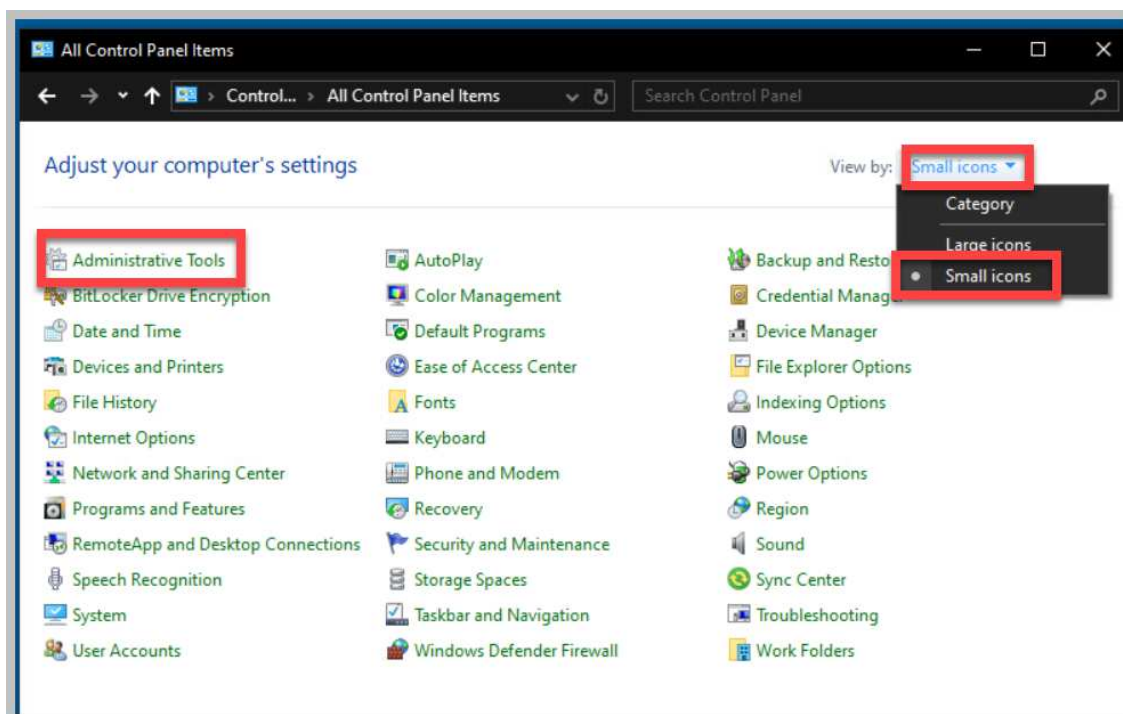


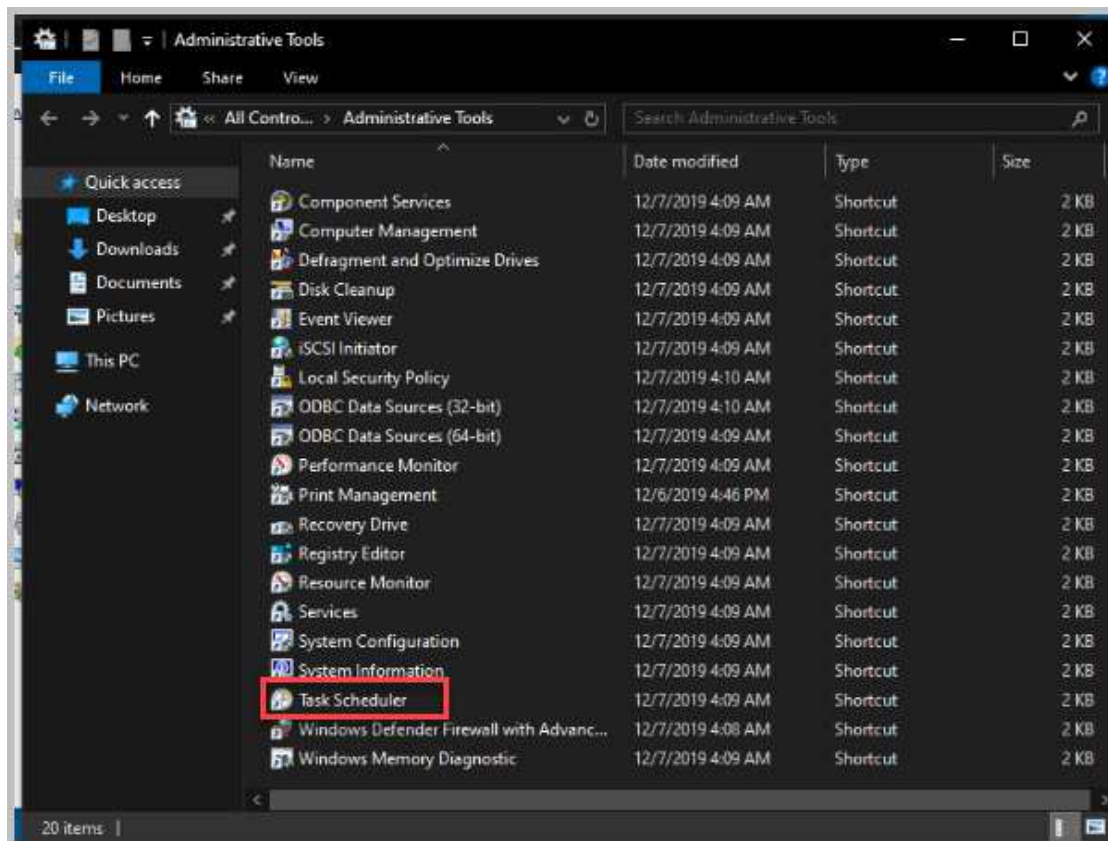
This process will create a set of registry keys that will store the cleanup settings you just have chosen. The value (1) you put in the command line allows you to store different cleanup options that can be run by the scheduler by using the `/sagerun:n` switch.

4. Select the search bar on the taskbar, type the command **Control Panel**, and select **Open**.

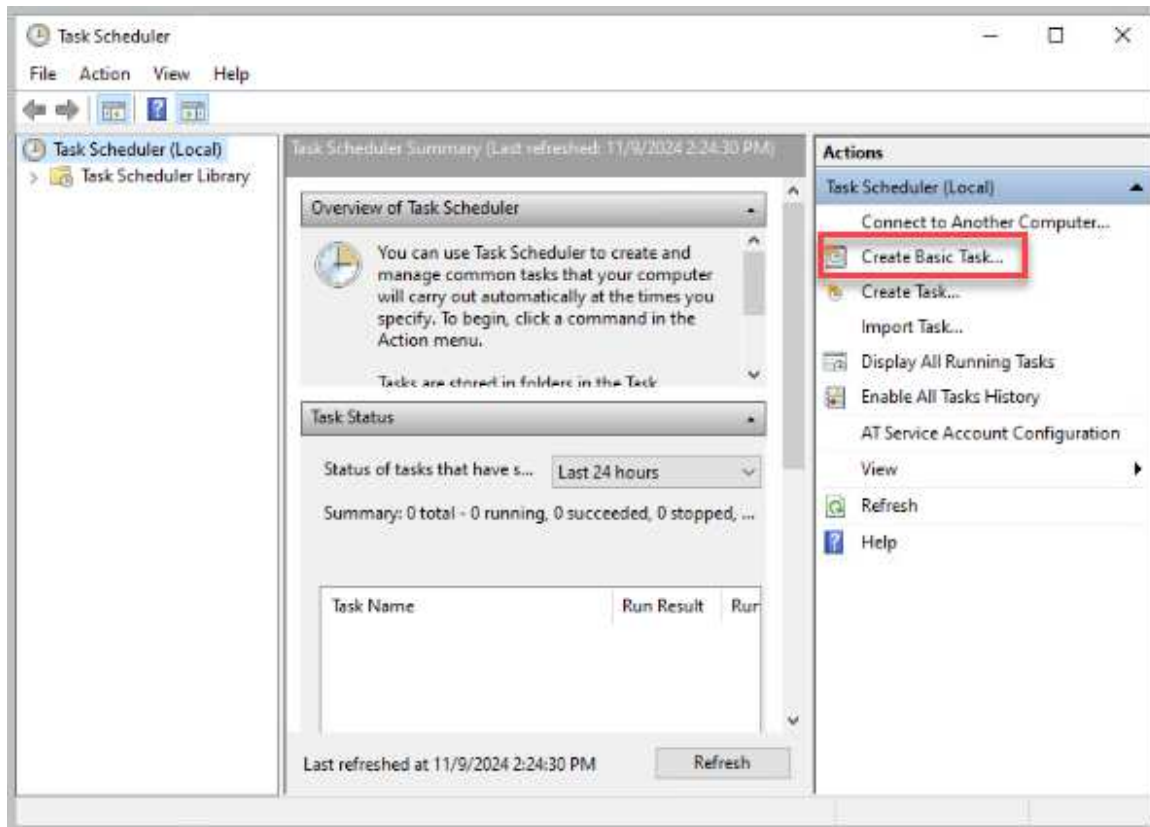


5. In the *Control Panel* window, change the *View* to **Small icons**. Click on **Administrative Tools**.

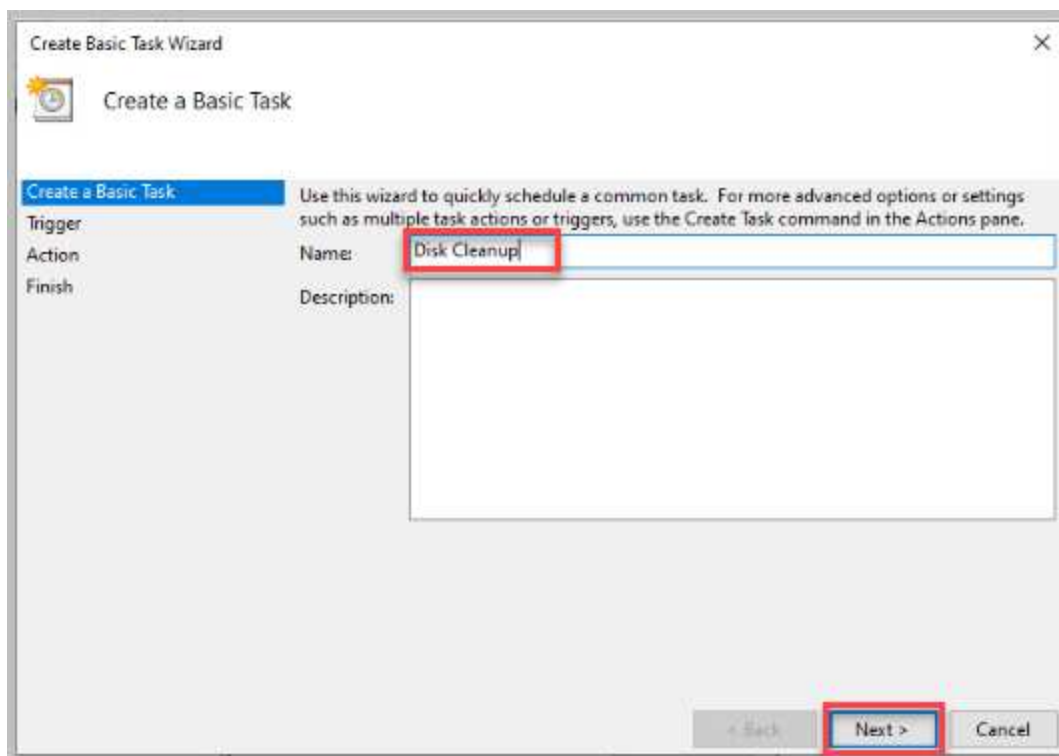


6. Double-click on **Task Scheduler**.

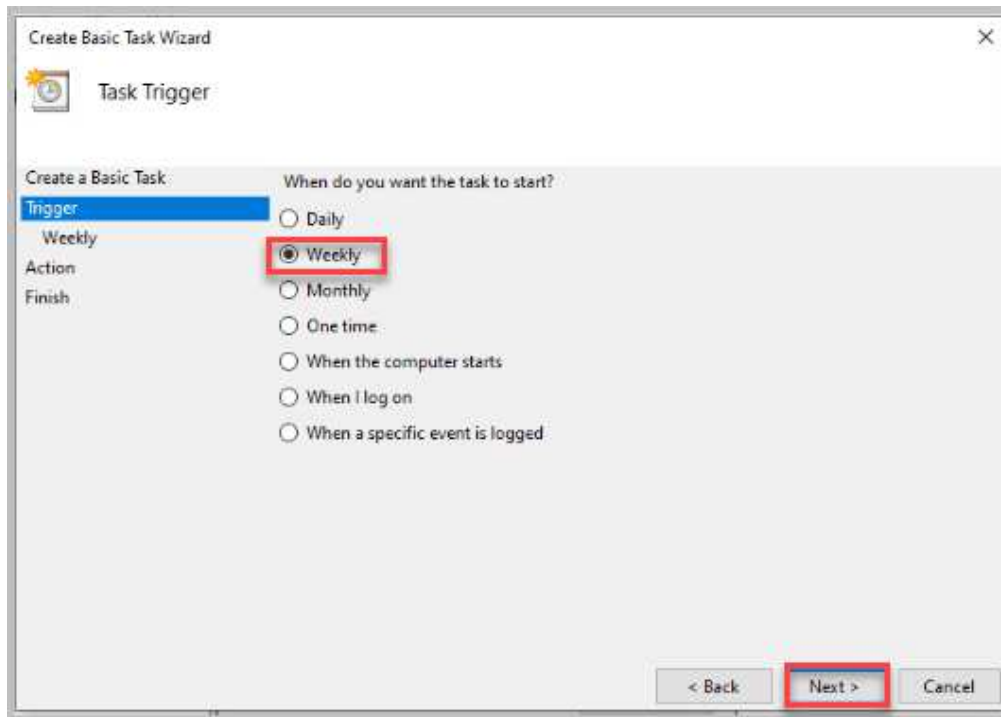
7. In the *Task Scheduler* window, on the right-side pane, underneath *Actions*, click on **Create Basic Task**.



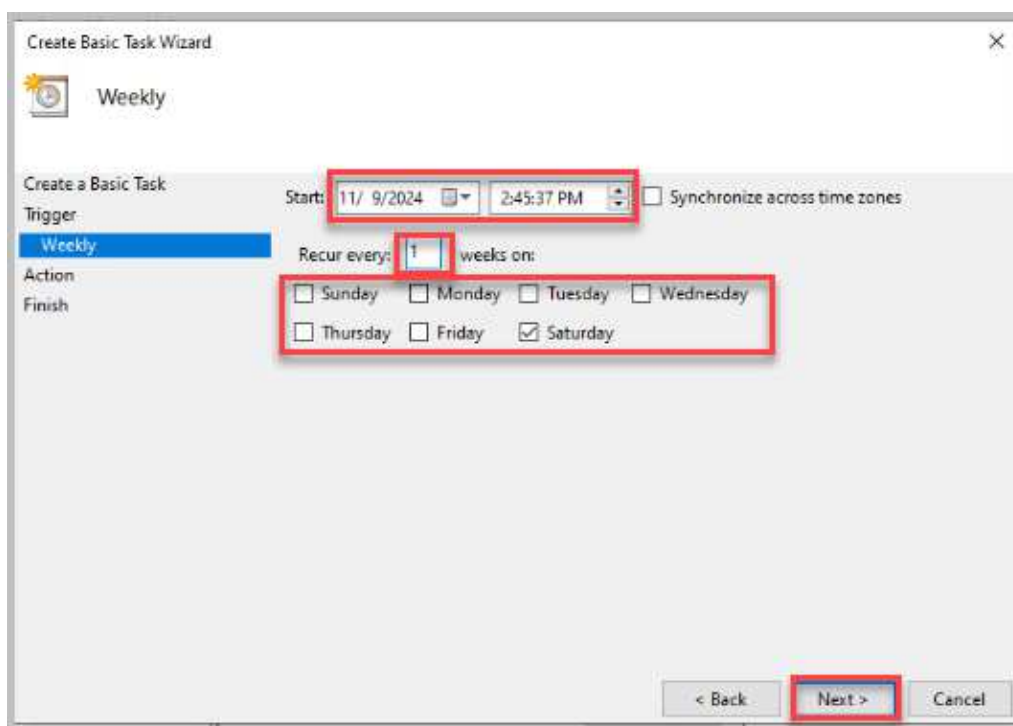
8. In the *Create Basic Task Wizard* window, type **Disk Cleanup** in the *Name* text field and click **Next**.



9. You will now be directed through a series of questions for scheduling options. First, you are asked when you want the task to start. Click the radio button next to **Weekly** and click **Next**.



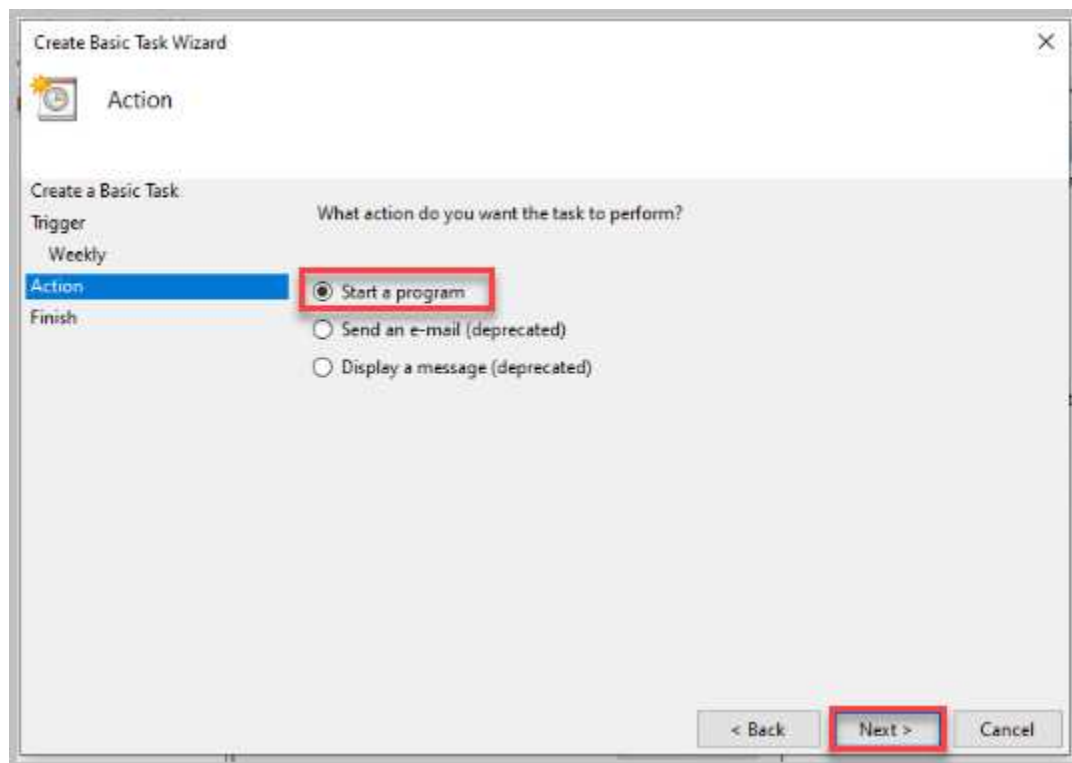
10. Next, specify the options for the scheduled task. Set the start time to **10-15 minutes** after your current time. This should give you enough time to finish the wizard before it initiates. Leave the value of **1** in the *Recur every* text field. Set the day of the week to your **current day**. Click the **Next** button.



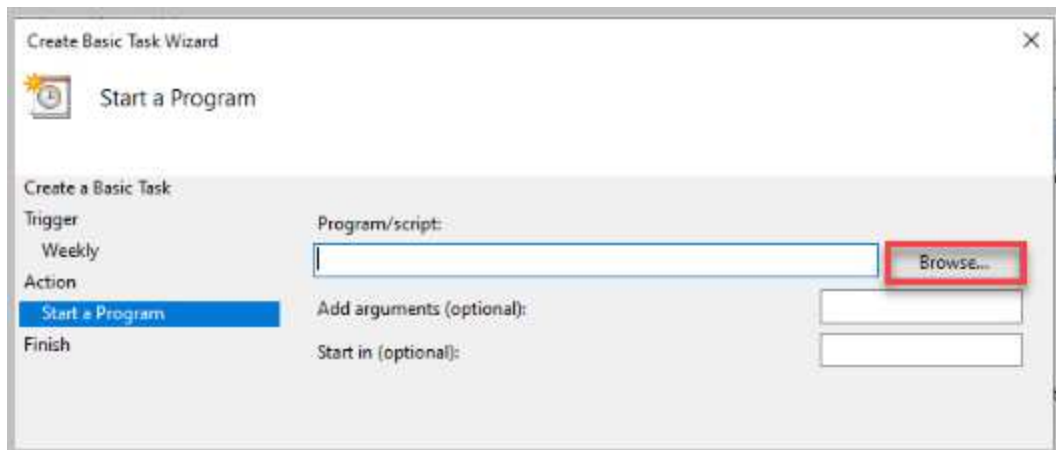


- *Start (date and time)* – this indicates when the scheduled task will be run for the first time. It is important to choose a starting time when the computer will not be in use. Even though the task will run in the background, it will slow the computer down and an unsuspecting user might think the computer is malfunctioning and reboot.
- *Recur every* – this indicates whether to run the task every week (1) every other week (2) every third week (3) and so on.
- *Sunday, Monday, Tuesday, etc. checkboxes* – this will say on

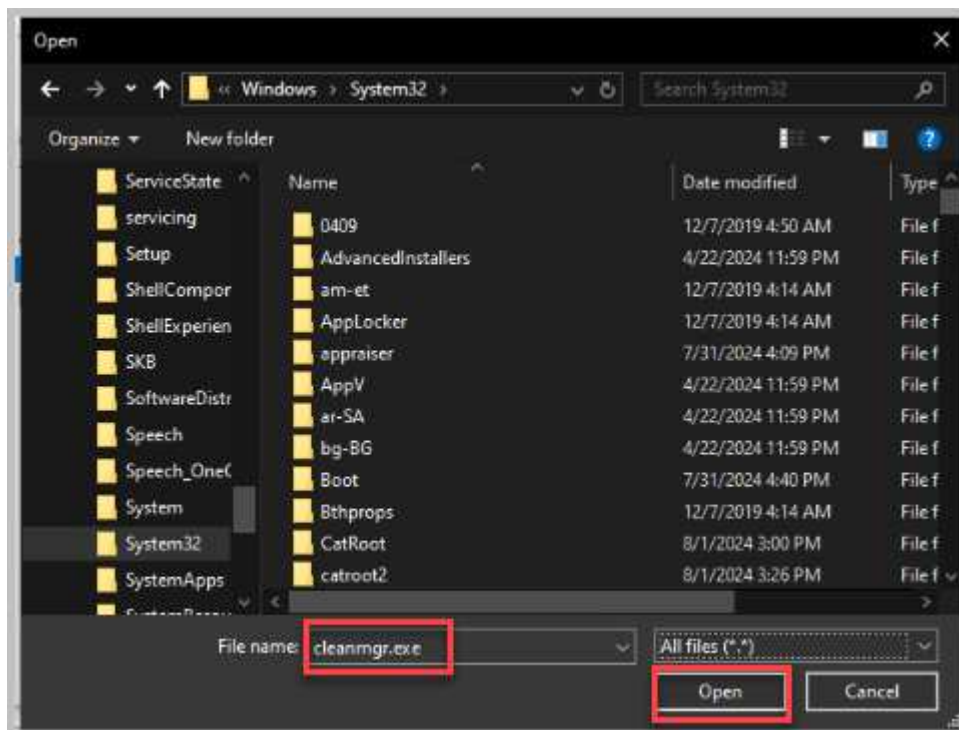
11. On the next step, ensure that **Start a program** is selected and click **Next**.



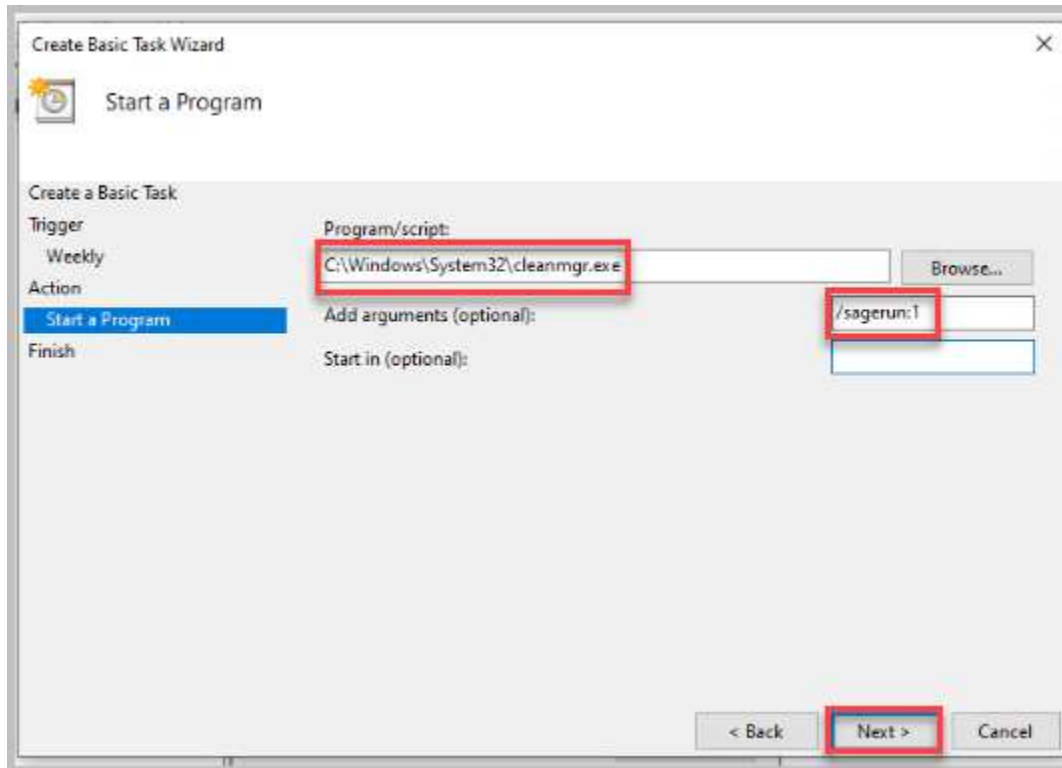
12. Click the **Browse** button for *Program/script*.



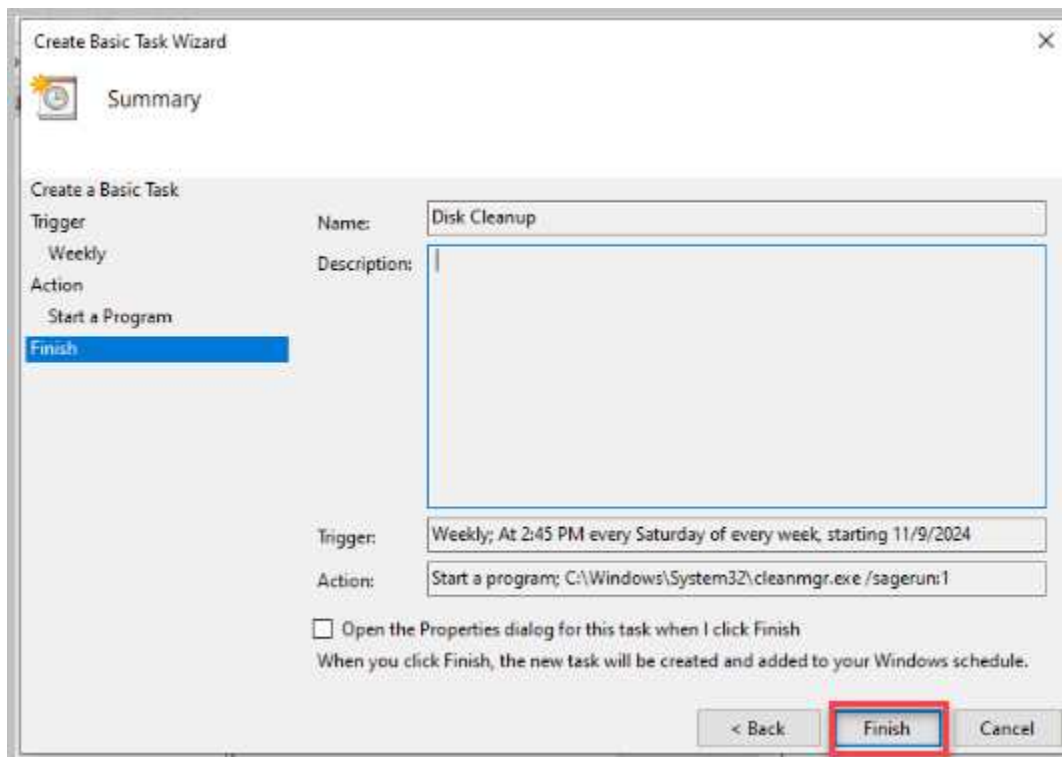
13. In the *Windows Explorer* window, ensure that you are in the *C:\Windows\System32* directory and type *cleanmgr.exe* in the *File name* text field. Click **Open**.



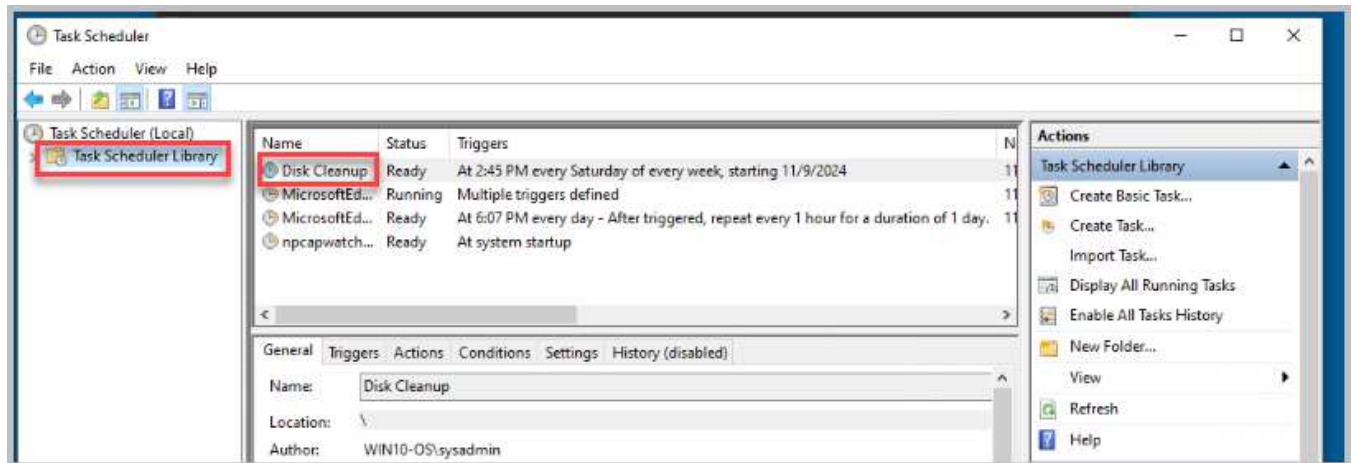
14. The program name will now display in the *Program/script* text field. Type `/sagerun:1` in the *Add arguments (optional)* text field. This will use the folder and area cleanup settings you configured earlier. Click **Next**.



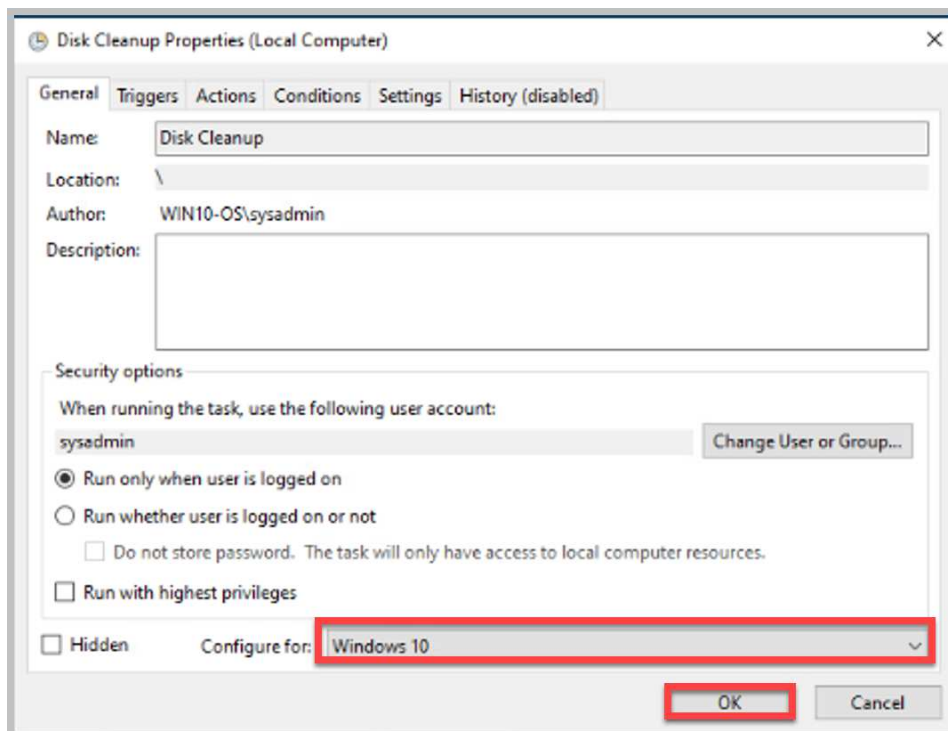
15. On the *Finish* page, confirm the settings and click **Finish**.



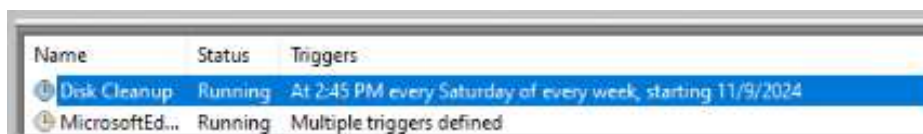
16. Back on the *Task Scheduler* window, in the left pane, click on the **Task Scheduler Library** folder. Notice the *Disk Cleanup* task is now listed in the center pane. Double-click on the **Disk Cleanup** task in the center pane.



17. In the *Disk Cleanup Properties* window, select the list arrow next to the *Configure for* option and change it to **Windows 10**. Click **OK**.



18. At this point, if enough time has passed for the scheduled task, you should see a status of *Running* for the *Disk Cleanup* task. If you miss the running status, you can check when the schedule was last run by looking at the *Last Run Time* column.



19. Close the **Task Scheduler** window.



Challenge Task:

As a challenge, try to recreate this task in a different version of Windows OS that is made available in this lab environment. This will help show similarities and differences between the versions.

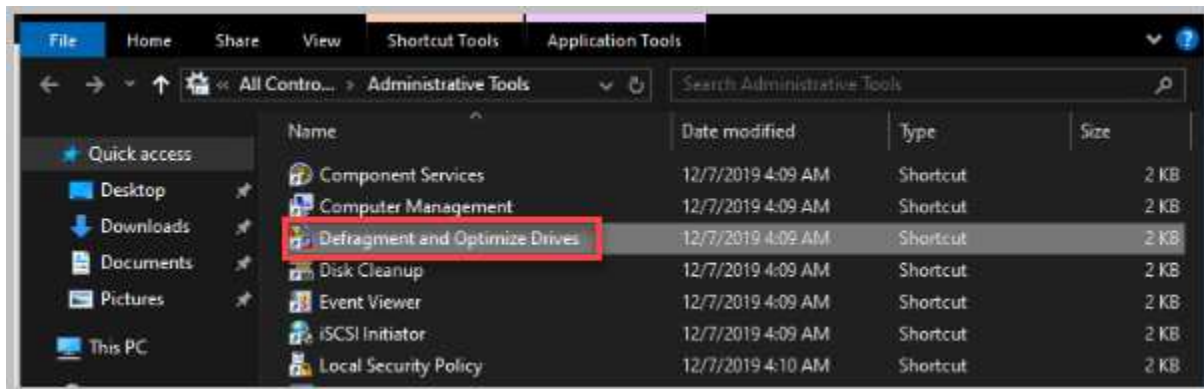
20. Leave the *Win10-OS* window open to continue with the next task. Leave the *Control Panel* and *Administrative Tools* window open.

2 Running the Disk Defragmentation Utility

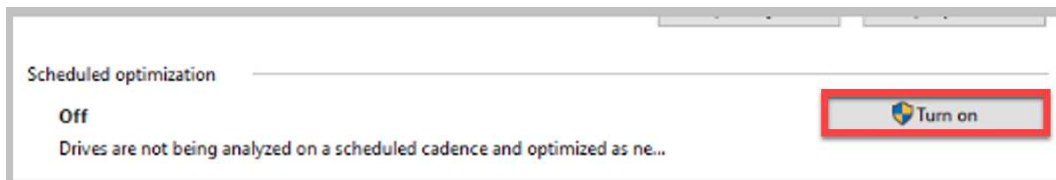
Deleting files (including temporary files) can cause a situation called *disk fragmentation*, which can increase read/write times on a hard drive, affecting performance. Defragmenting is recommended as part of a good maintenance plan.

2.1 Running Disk Defragmentation on Win10-OS

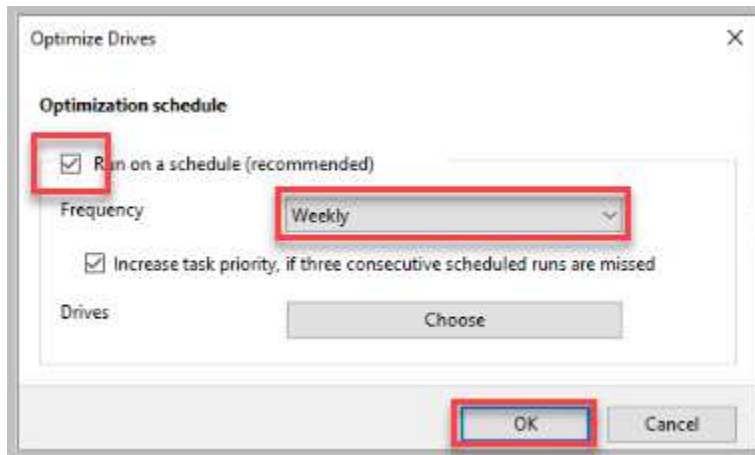
1. On the *Win10-OS* machine, with *Administrative Tools* opened, double-click on **Defragment and Optimize Drives**.



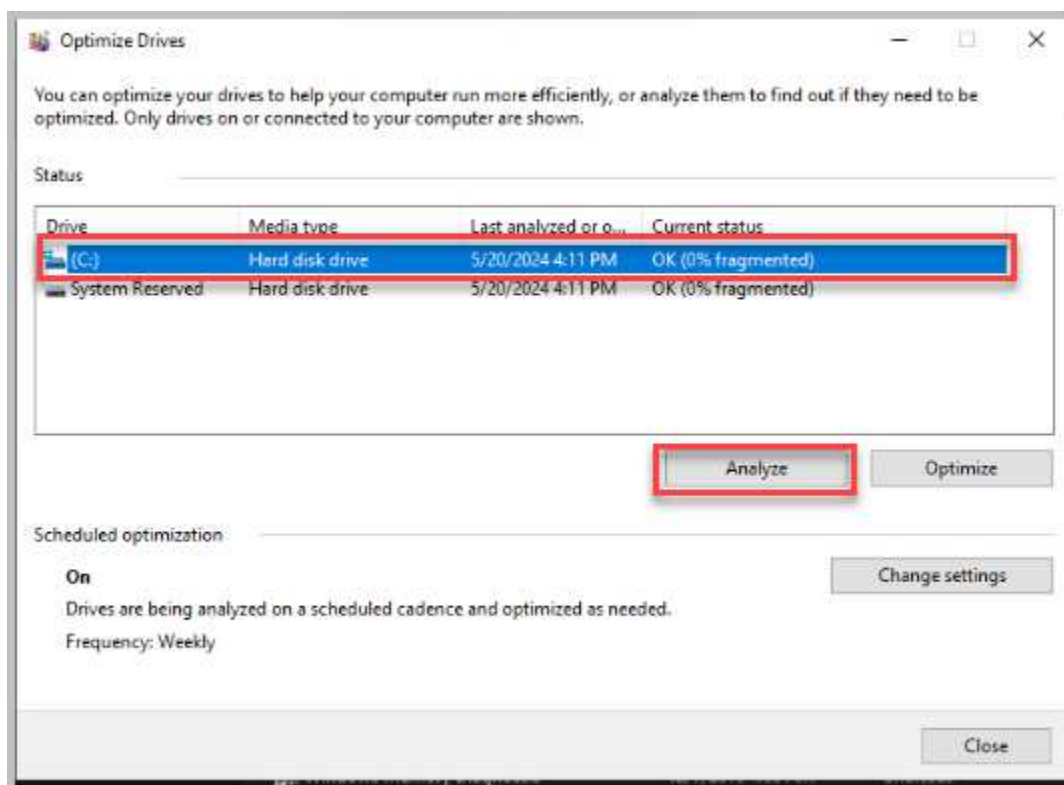
2. To change the scheduled defragmentation and optimization times, click the **Turn on** button in the *Scheduled optimization* pane.



3. Notice that a pop-up window appears. Check the checkbox for **Run on a schedule (recommended)**. Select **Weekly** from the *Frequency* dropdown menu and click **OK**.

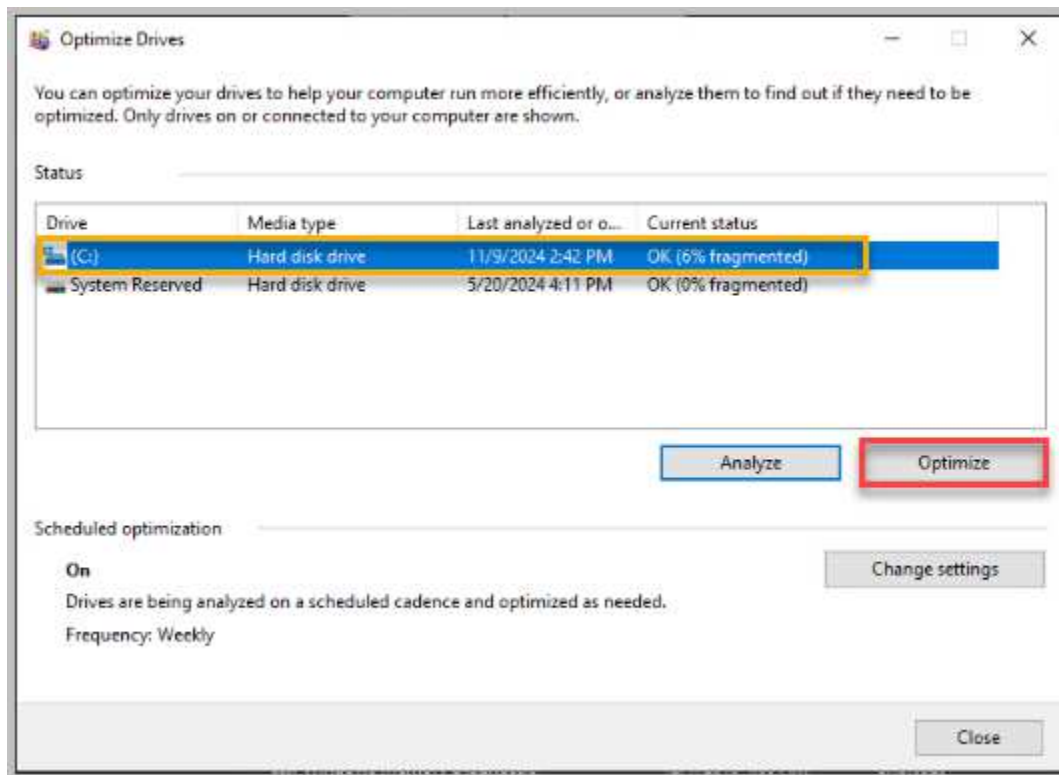


4. Initiate the *Disk Defragmentation* utility by first analyzing the disk. Select **(C:)** from the list of drives and click the **Analyze** button.



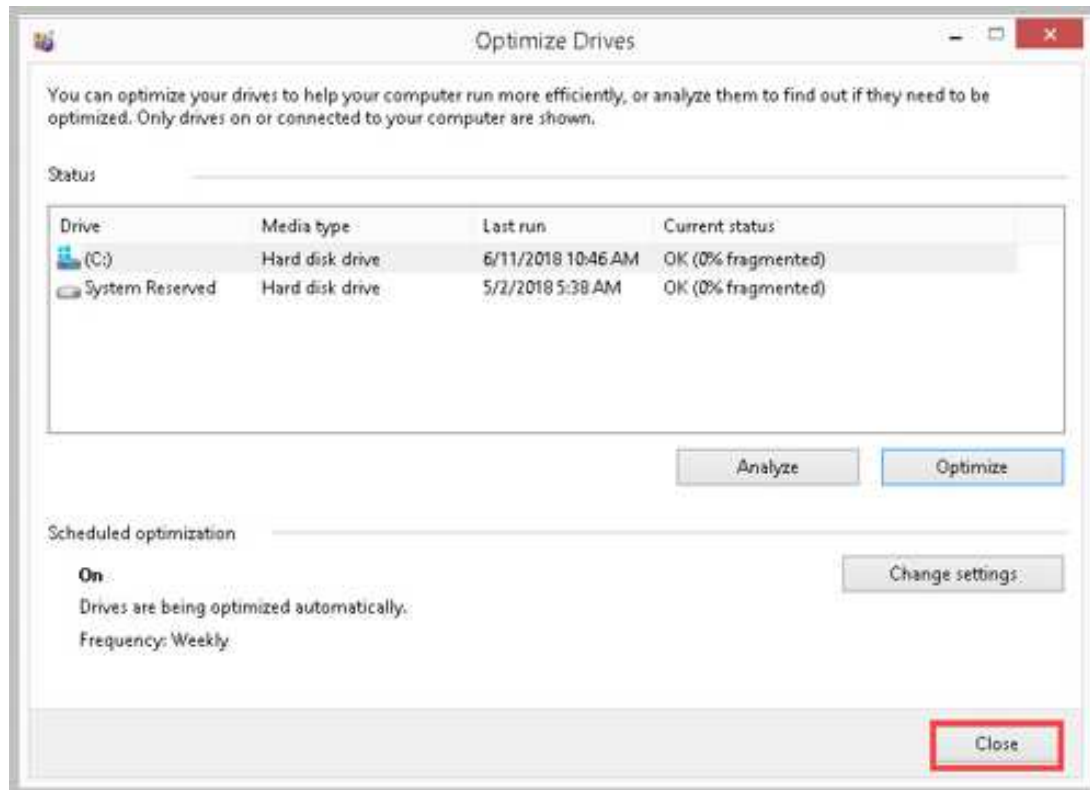
The disk analysis should take about 1-2 minutes.

5. Once the disk analysis completes, notice the percentage of fragmentation for C: underneath the *Current status* column. In this example, 6% of the disk is fragmented, meaning that 6% of files are not placed next to each other on the hard drive. Proceed by ensuring that (C:) is highlighted, then click the **Optimize** button. Once complete, the Lab output shows OK (100% space efficiency).



The disk defragmentation for this virtual machine should take about 3-4 minutes.

6. Once the defragmentation completes, notice the percentage underneath the *Current status* column. Click **Close** when finished.



Challenge Task:

As a challenge, try to recreate this task in a different version of Windows OS that is made available in this lab environment. This will help show similarities and differences between the versions.

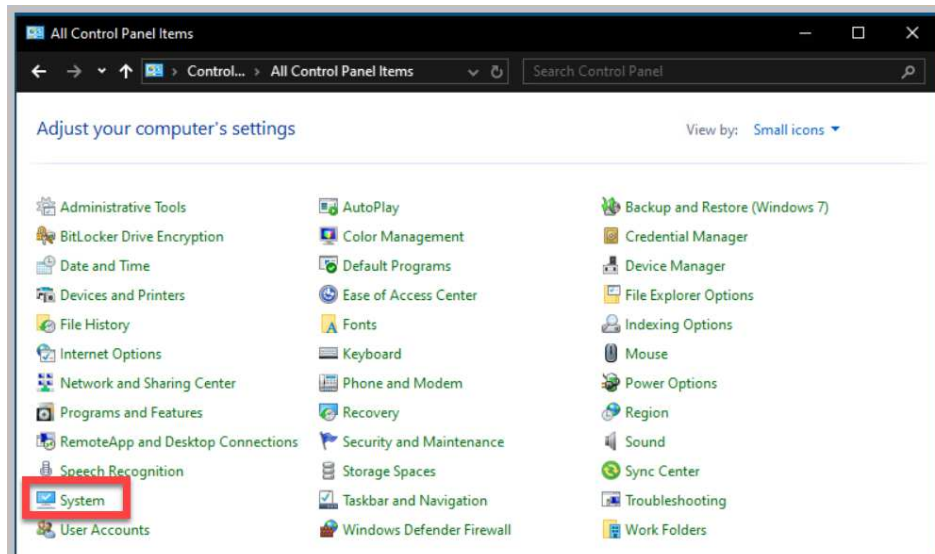
7. Close the **Administrative Tools** window.
8. Leave the *Win10-OS* window open to continue with the next task.

3 Creating System Restore Points

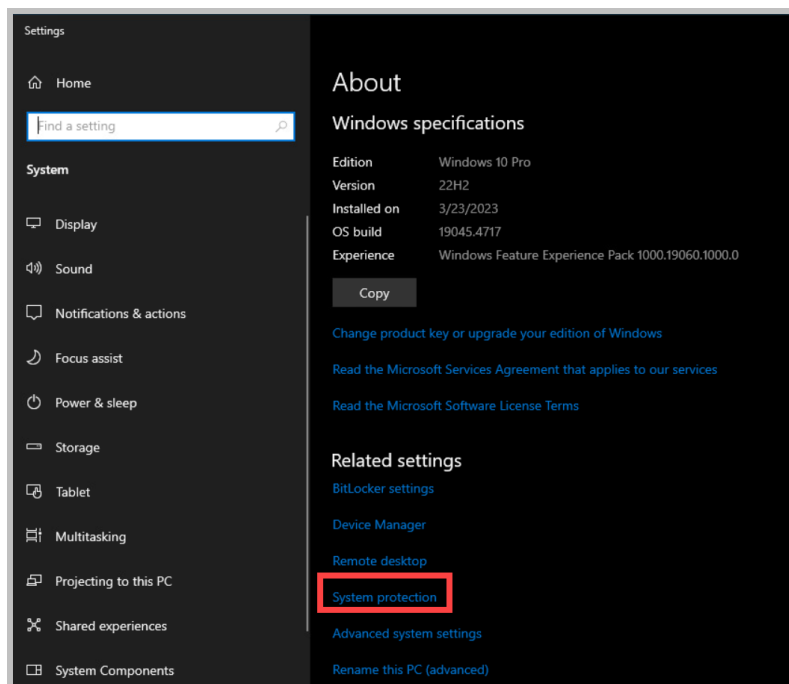
The *System Restore* function lets users return the computer to a previous state, including all system files, installed applications, system settings, and the *Windows Registry*. It can be beneficial when a system-level problem occurs.

3.1 Initiating a System Restore on Win10-OS

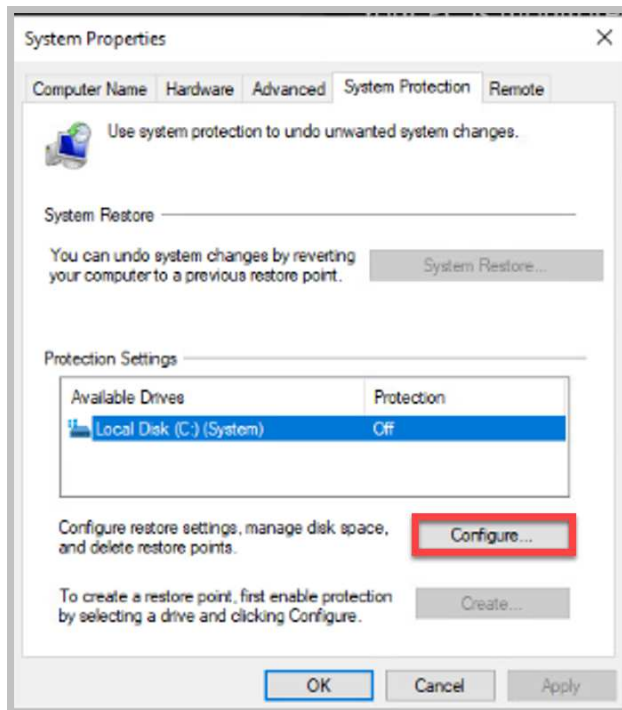
1. While on the *Win10-OS* system, in the *Control Panel* window, click on the **System** applet.



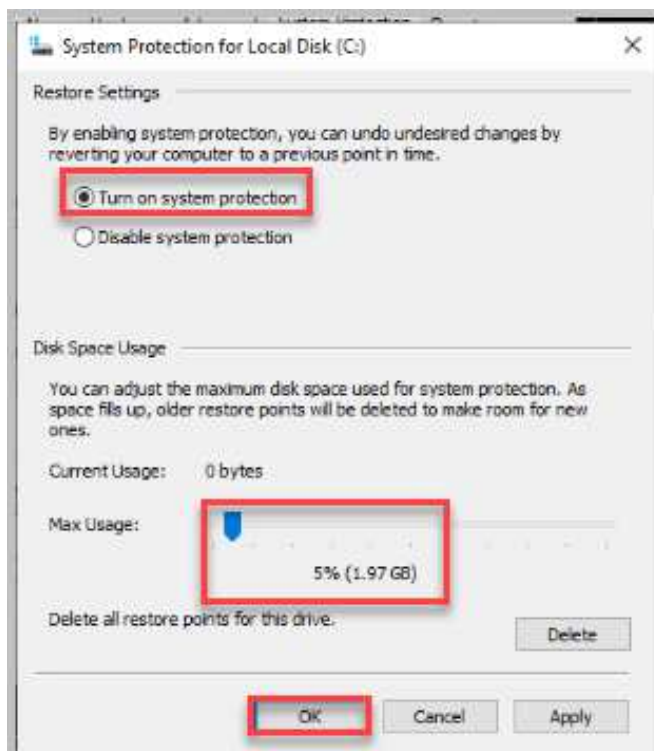
2. Click on **System protection** from the center pane.



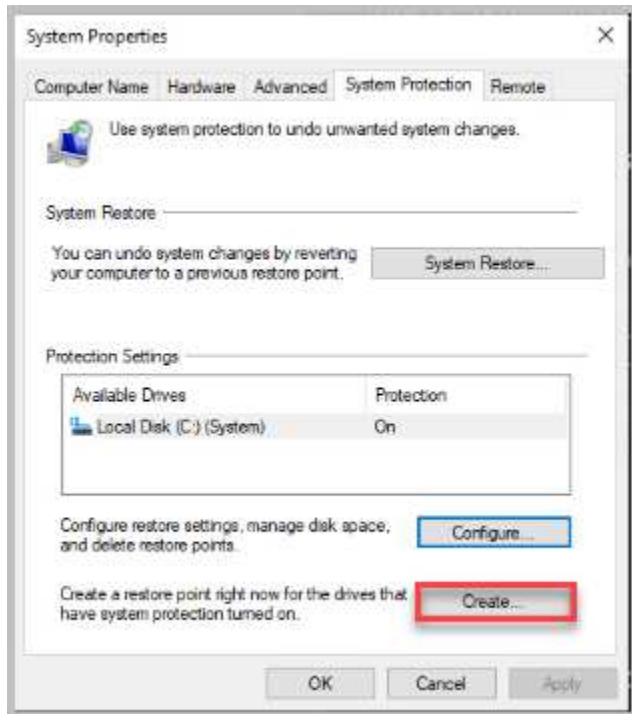
3. In the *System Properties* window, on the *System Protection* tab, click the **Configure** button to configure restore settings, manage disk space, and delete restore points.



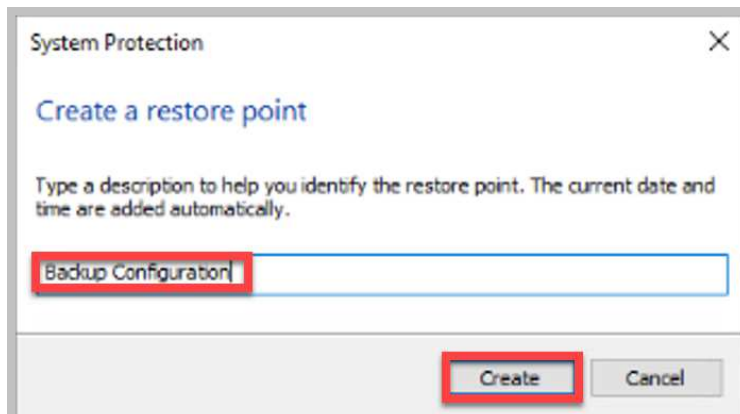
4. In the *System Protection for Local Disk (C:)*, select **Turn on system protection**. In the *Disk Space Usage*, you can adjust the maximum disk space used for system protection. In a normal setting, you would set a higher amount of space to use to keep a higher number of restore points. For this lab activity, set the *Max Usage* to **5% (1.97GB)**. Click **OK** to apply and close the setup for System Protection.



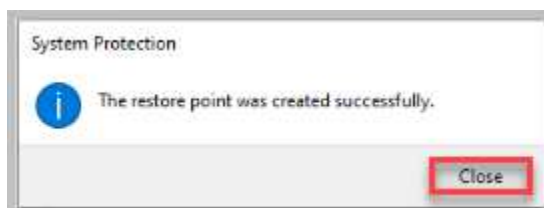
5. In the *System Properties* window, on the *System Protection* tab, click the **Create** button to begin creating a restore point.



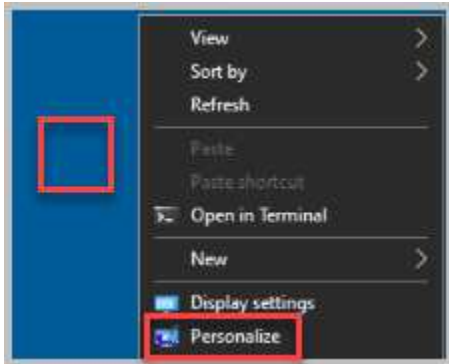
6. When prompted to enter a description for the restore point, type **Backup Configuration** in the text field and click the **Create** button.



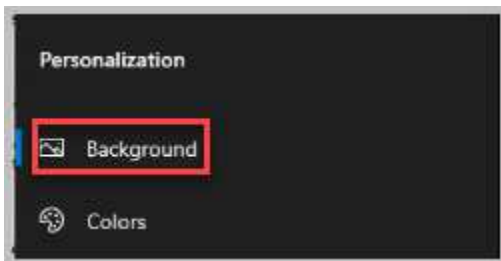
7. Notice the utility will create the restore point. Once finished, click the **Close** button.



8. Now, change the background to test the restore point. Right-click on the desktop and click **Personalize**.



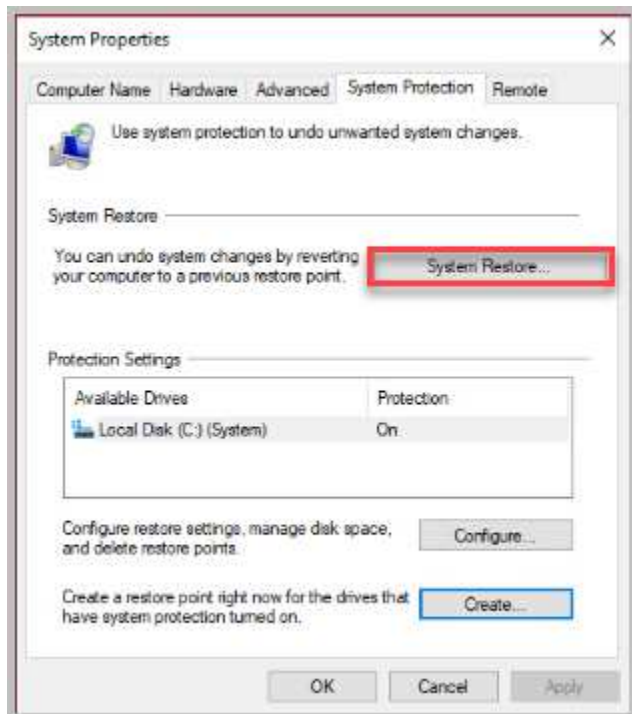
9. In the *Personalization* window, click on the **Background** link.



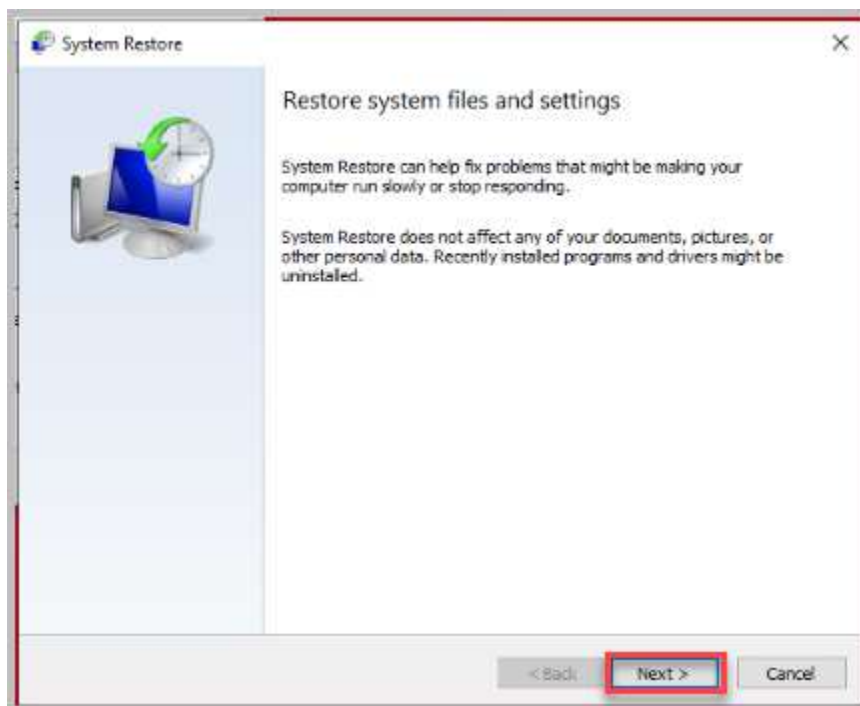
10. Ensure that **Solid color** is selected from the dropdown menu for *Picture location*, and select any color that is different from the current background color. Close the **Settings** window.



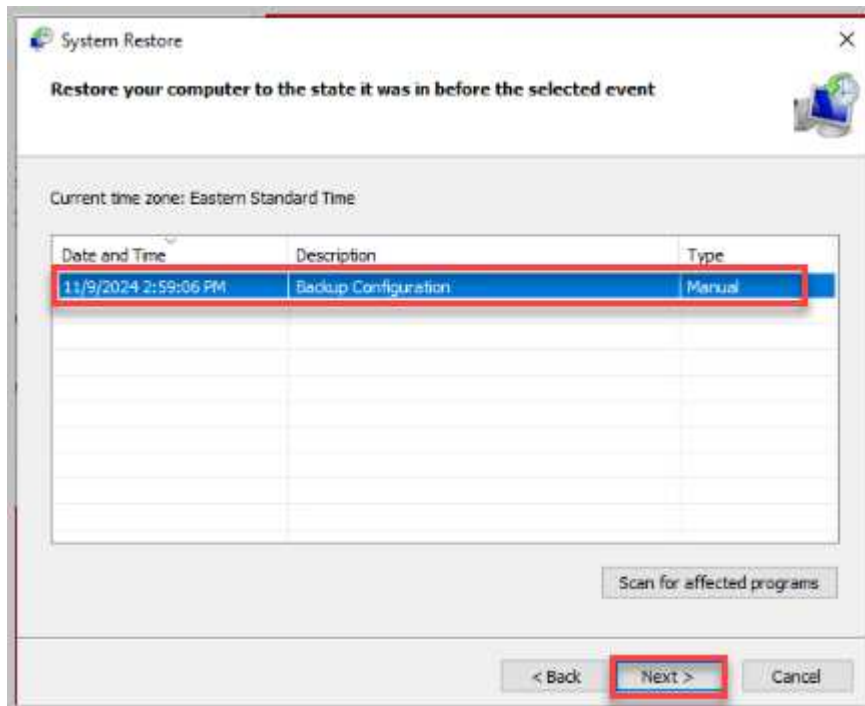
11. Change focus to the **System Properties** window, while on the *System Protection* tab, click on the **System Restore** button.



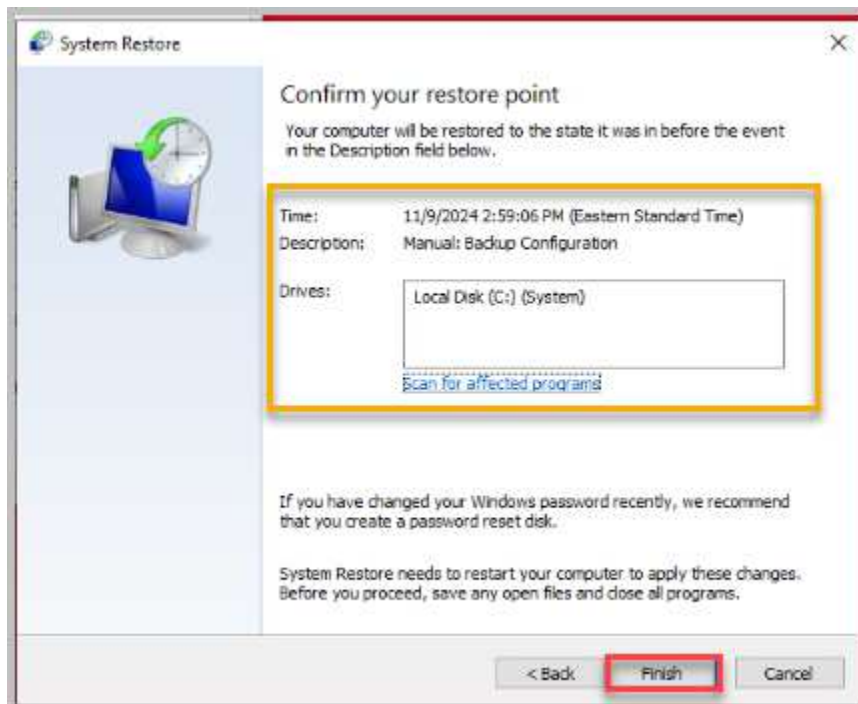
12. Notice that the *System Restore* wizard appears. Click **Next**.



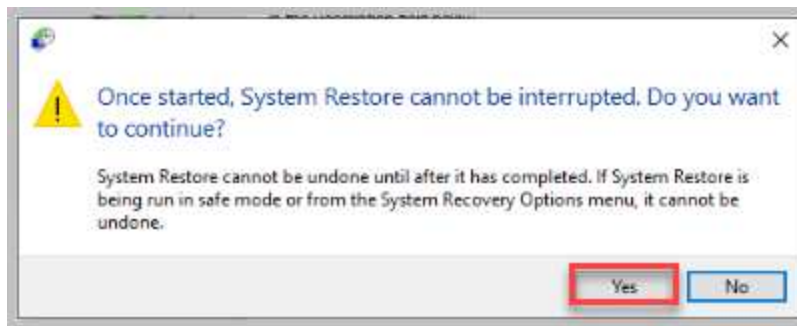
13. A list of all the recent restore points will be shown. Select the **Backup Configuration** restore point and then click **Next**.



14. Verify the restore point information and click the **Finish** button.

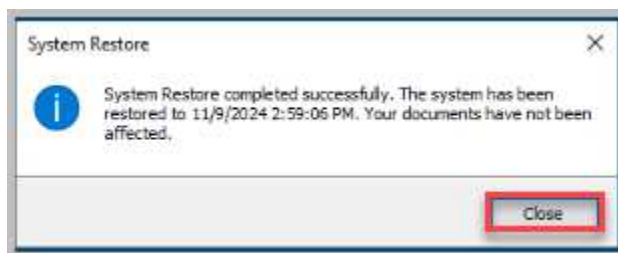


15. When prompted, click **Yes** to continue.



The system restore process should take about 8 minutes.

16. Once the system restore point has been completed, the Windows OS will reboot itself and log in automatically.
17. When logged in, notice the desktop background has reverted to its original settings. On the *System Restore* confirmation window, click the **Close** button. Leave the *Win10-OS* system open for the next activity.



Challenge Task:

As a challenge, try to recreate this task in a different version of Windows OS that is made available in this lab environment. This will help show similarities and differences between the versions.

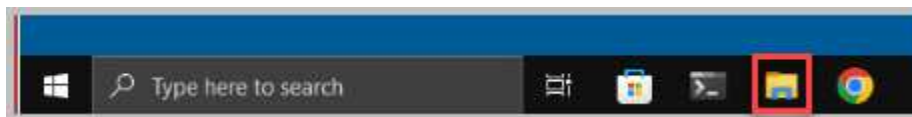
4 Windows Backup

Backups are a function that needs to be performed regularly (for corporate computers, daily at least, and for individuals, minimally once a week and more often if possible). When the computer fails, the only way to get your data back is with a good backup.

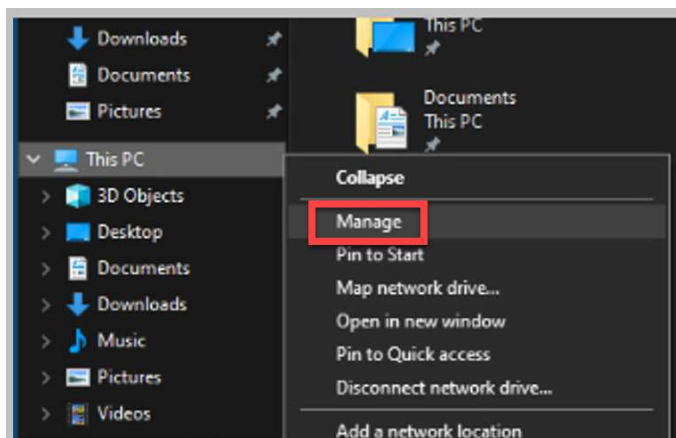
Fortunately, Windows includes a workable backup solution.

4.1 Setting up a Backup Location on Win10-OS

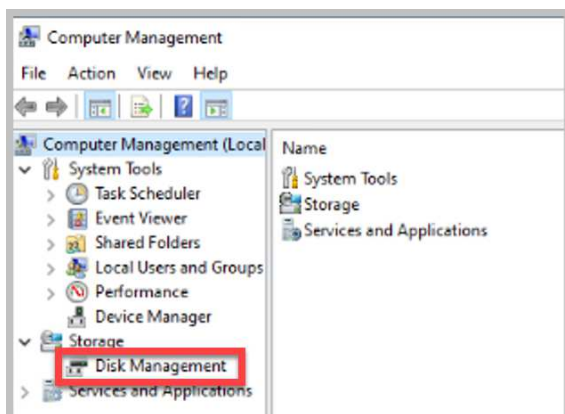
1. Before setting up a Backup Process with the *Win10-OS System*, you will need to set up a new Backup Location.
2. Select **File Explorer** on the taskbar.



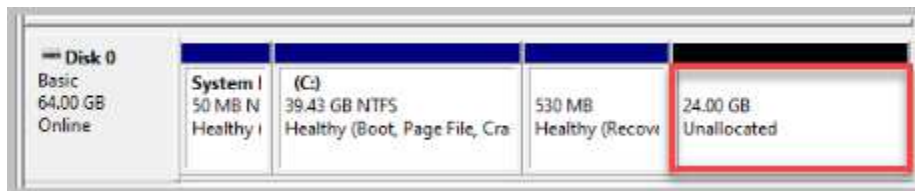
3. Right-click on the **This PC** icon and select **Manage**.



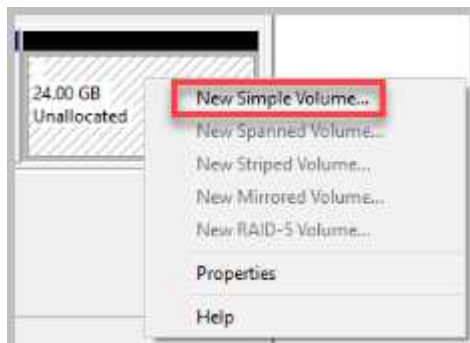
4. In the *Computer Management* window, click on **Disk Management** located in the left pane.



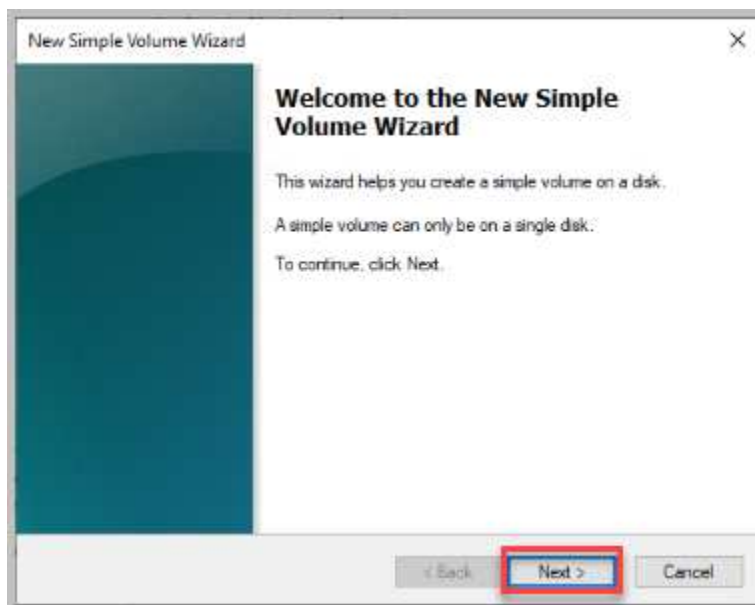
5. On *Disk0*, right-click the **(24 GB) Unallocated** space of the drive.



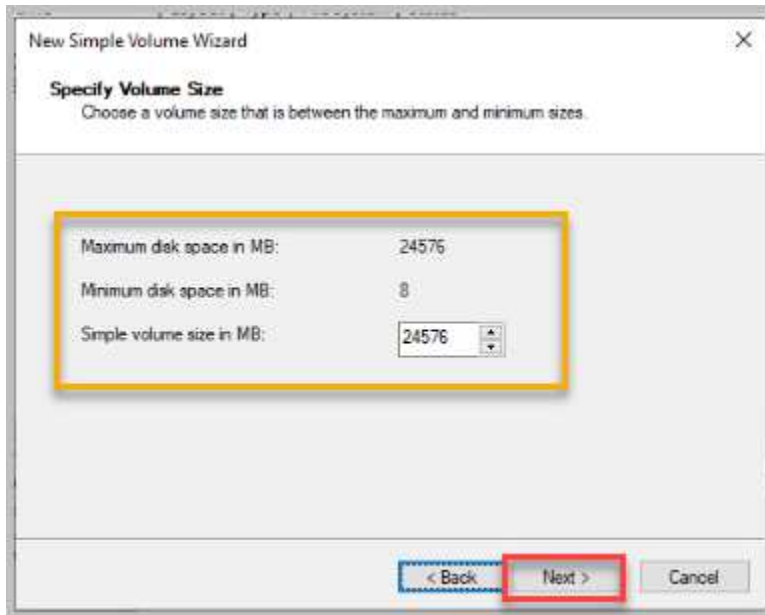
6. Select **New Simple Volume**.



7. In the *New Simple Volume Wizard*, select **Next**.

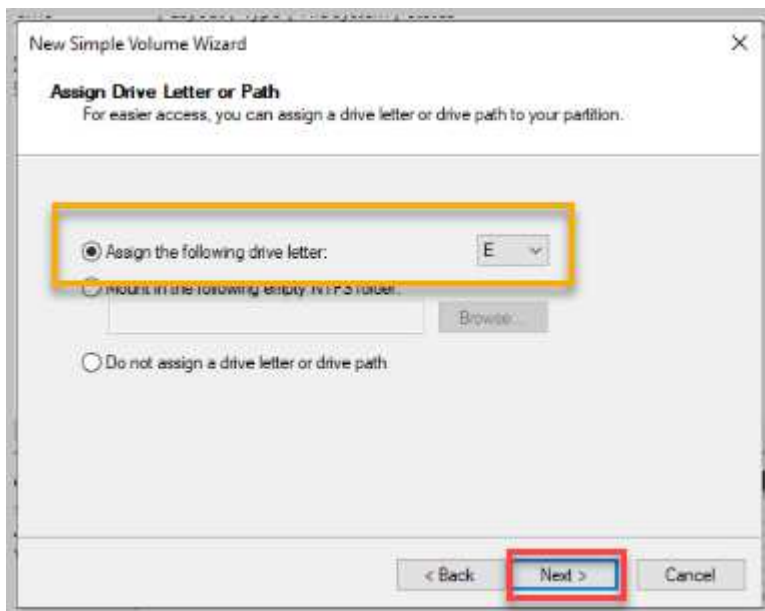


8. Leave the *Default Settings*, and select **Next**.



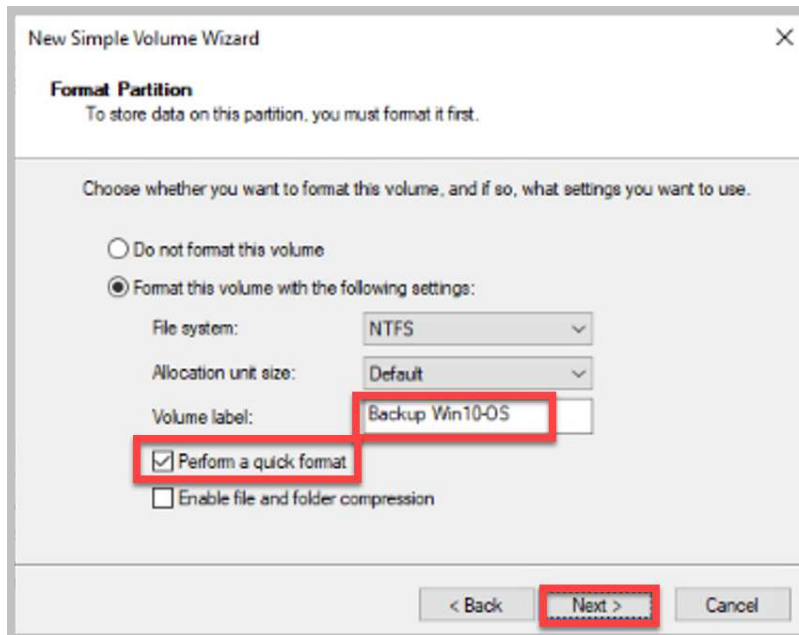
The screenshot shows the 'New Simple Volume Wizard' window at the 'Specify Volume Size' step. The window title is 'New Simple Volume Wizard' with a close button (X) in the top right corner. Below the title bar, the text 'Specify Volume Size' is followed by the instruction 'Choose a volume size that is between the maximum and minimum sizes.' The main area contains three labels and their corresponding values: 'Maximum disk space in MB:' with the value '24576', 'Minimum disk space in MB:' with the value '8', and 'Simple volume size in MB:' with a text box containing '24576' and a small up/down arrow. At the bottom of the window, there are three buttons: '< Back' (highlighted with a blue dashed border), 'Next >' (highlighted with a red solid border), and 'Cancel'.

9. Leave the default drive letter assigned to **E** and select **Next**.

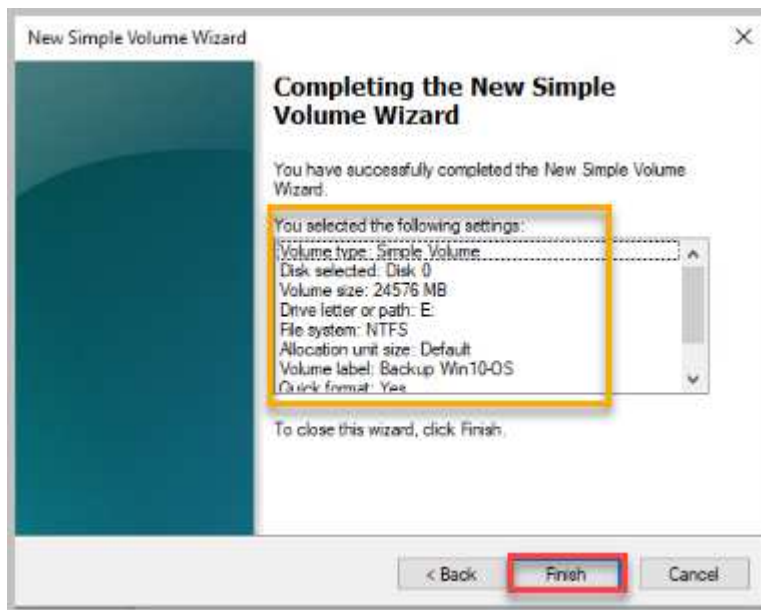


The screenshot shows the 'New Simple Volume Wizard' window at the 'Assign Drive Letter or Path' step. The window title is 'New Simple Volume Wizard' with a close button (X) in the top right corner. Below the title bar, the text 'Assign Drive Letter or Path' is followed by the instruction 'For easier access, you can assign a drive letter or drive path to your partition.' The main area contains three radio button options: 'Assign the following drive letter:' (selected, highlighted with a yellow border) with a dropdown menu showing 'E', 'Mount in the following empty NTFS volume:' (unselected), and 'Do not assign a drive letter or drive path' (unselected). Below the 'Mount in the following empty NTFS volume:' option is a text box and a 'Browse...' button. At the bottom of the window, there are three buttons: '< Back' (highlighted with a blue dashed border), 'Next >' (highlighted with a red solid border), and 'Cancel'.

10. Type **Backup Win10-OS** for the Volume Label and verify that the *Perform a quick format* has been checked. Select **Next**.



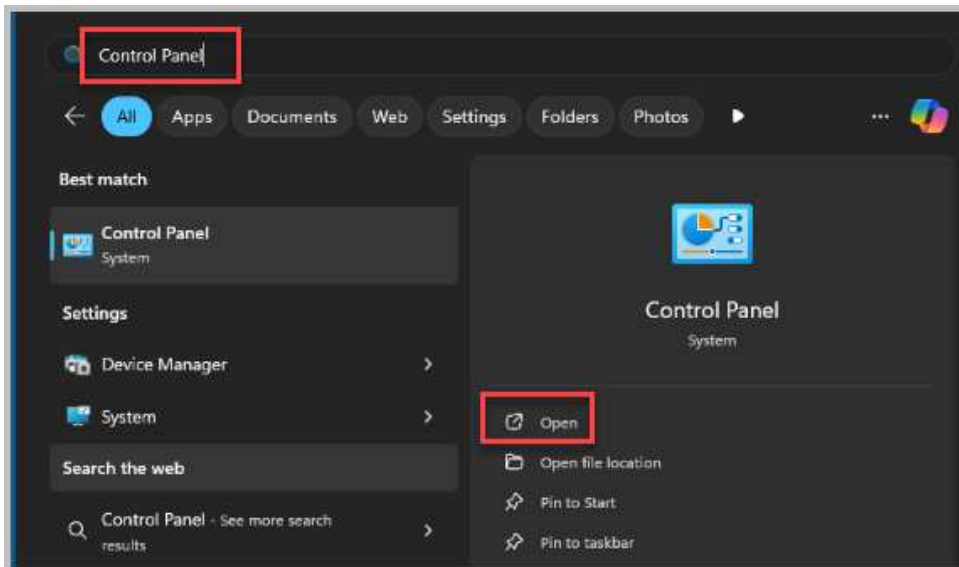
11. Verify the *New Simple Volume Settings* are correct and select **Finish**.



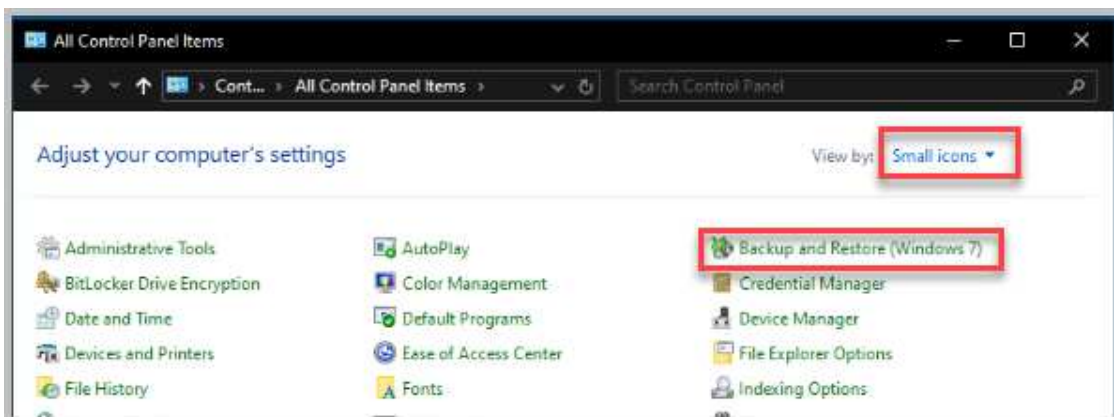
12. Close all open windows and leave the *Win10-OS system* open for the next lab activity.

4.2 Initiating a Backup Process on Win10-OS

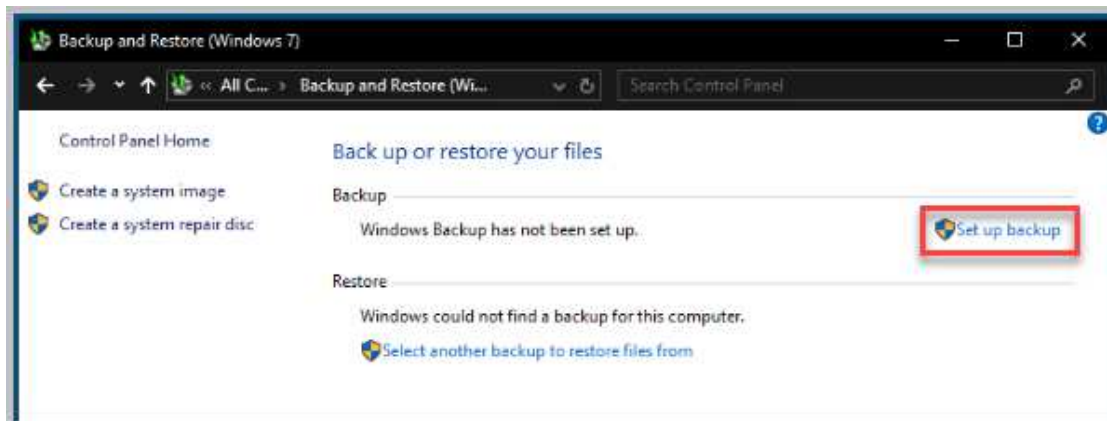
1. Select the search bar on the taskbar, type the command **Control Panel**, and select **Open**.



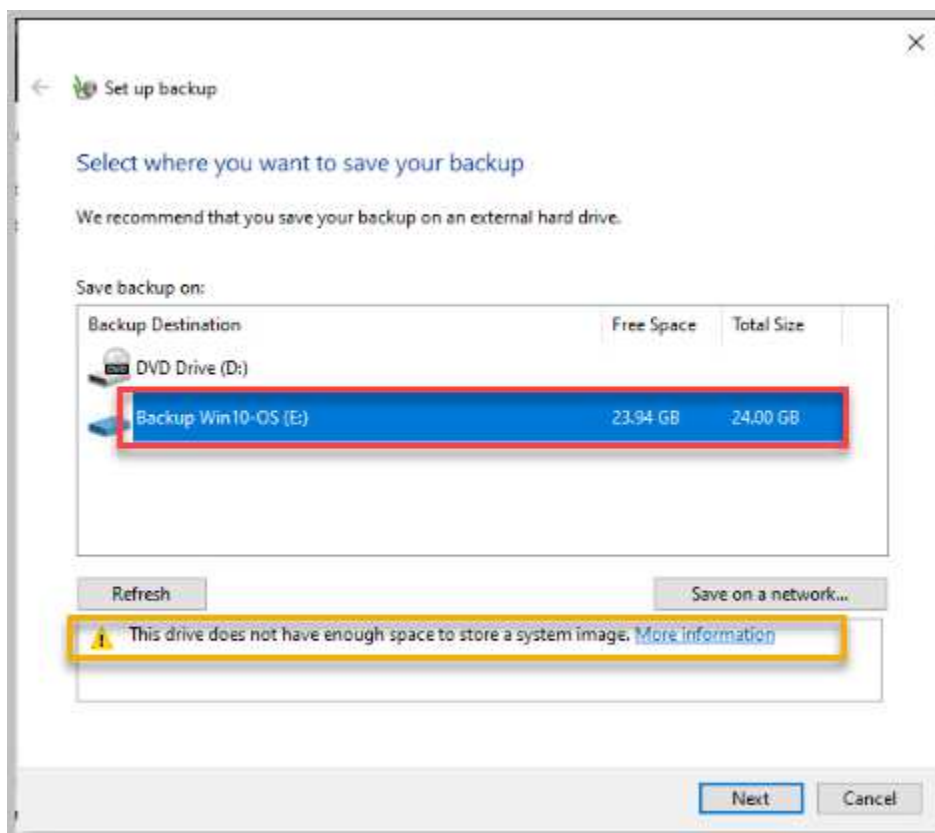
2. In the *Control Panel* window, change the *View* to **Small icons**. Click on **Backup and Restore (Windows 7)**.



- The first thing that will need to be done is to configure the backup settings. Click on the **Set up backup** link.



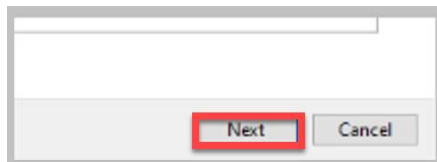
- For *Backup Destination*, select **Backup Win10-OS (E:)**. Notice the error message that this drive does not have enough space to store a system image.



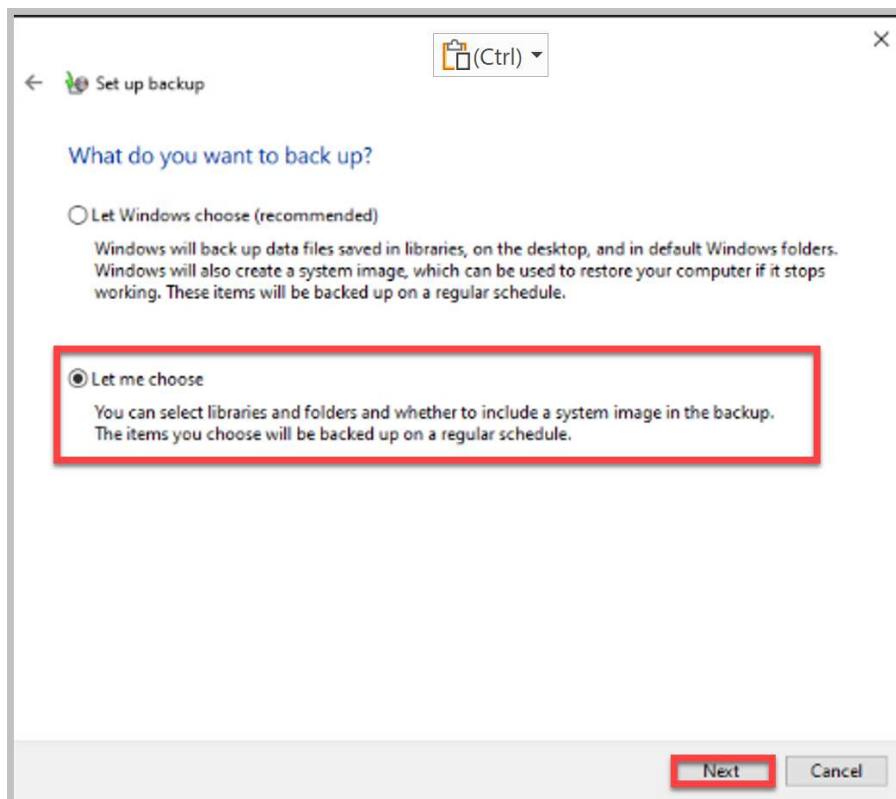
5. Select **More Information** from the error message. At a minimum, you need 33.13 GB of Free Space to complete a full backup of the *Win10-OS* System. Click **OK**.



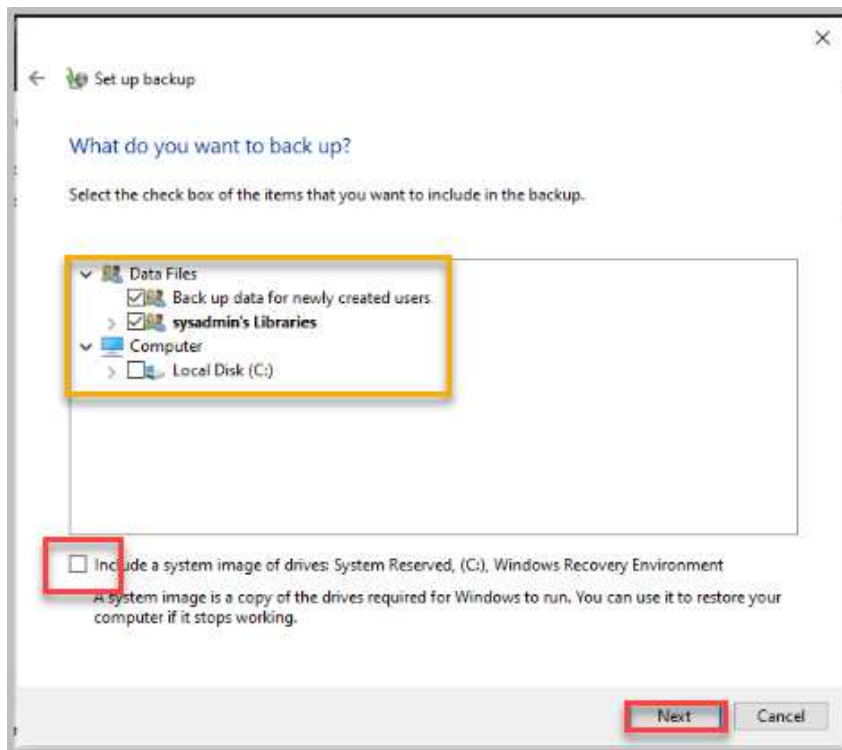
6. Select **Next**.



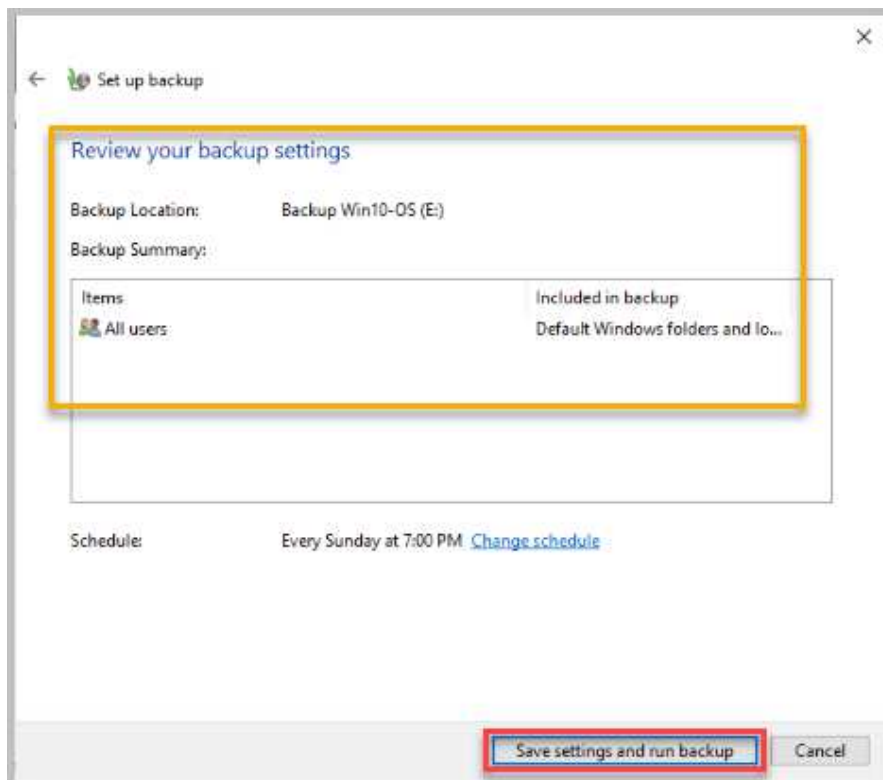
7. Select the option, **Let me choose** what to back up. Select **Next**.



8. Leave the default settings for *Data Files* and **Uncheck**. Include a system image and select **Next**.



9. Verify the items you are backing up and select **Save settings and run backup**.

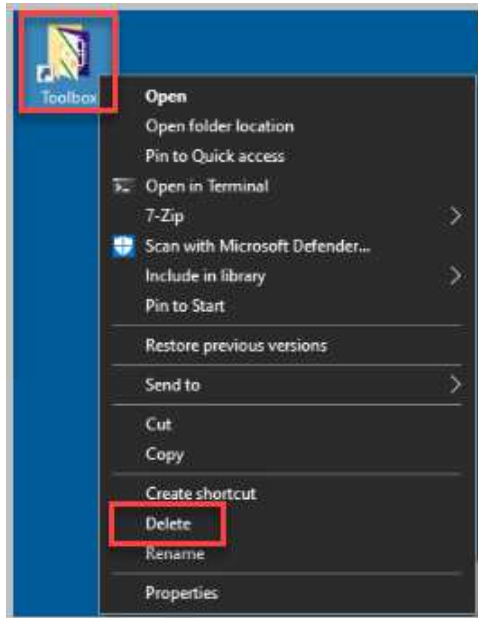


10. Notice that the backup will run for the first time and will continue to run according to the schedule that has been set.



The backup process should take about 4 minutes.

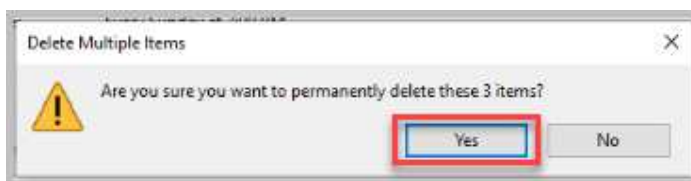
11. Now, to test the backup. There is a folder named *Toolbox* on the desktop that contains several programs. Right-click the **Toolbox** shortcut and click **Delete**.



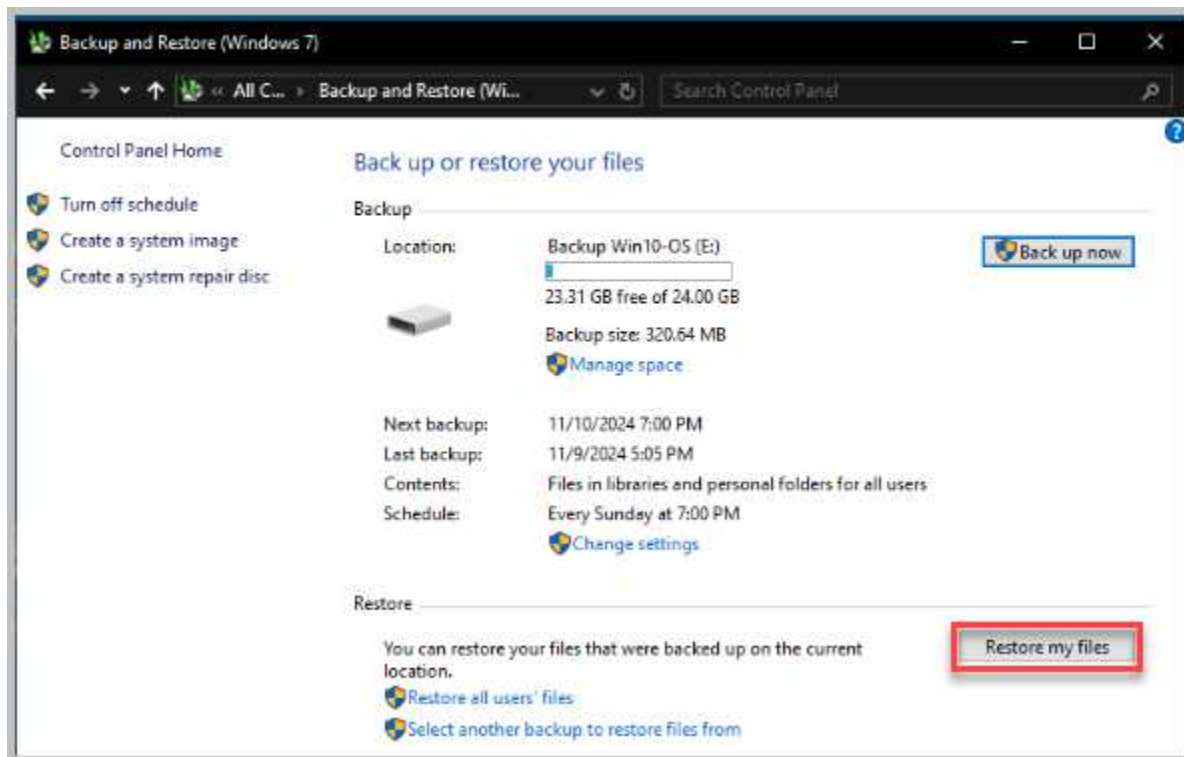
12. Right-click the **Recycle Bin** and select **Empty Recycle Bin**.



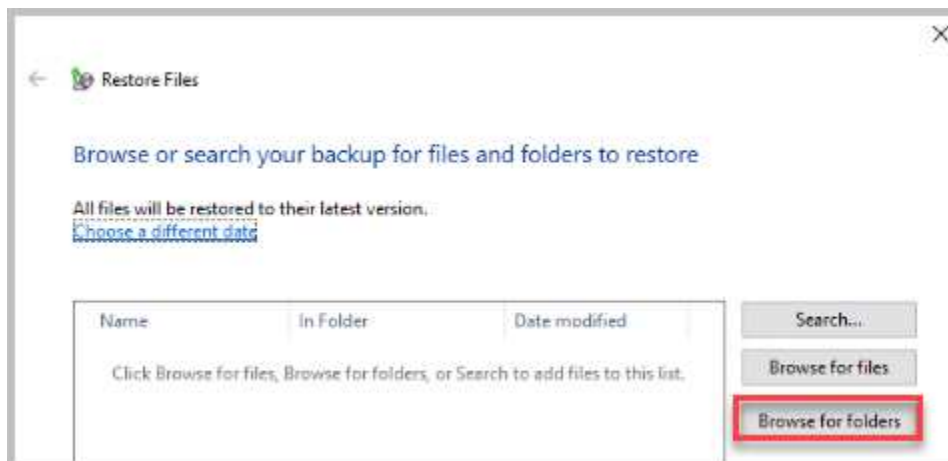
13. When prompted, click **Yes**.



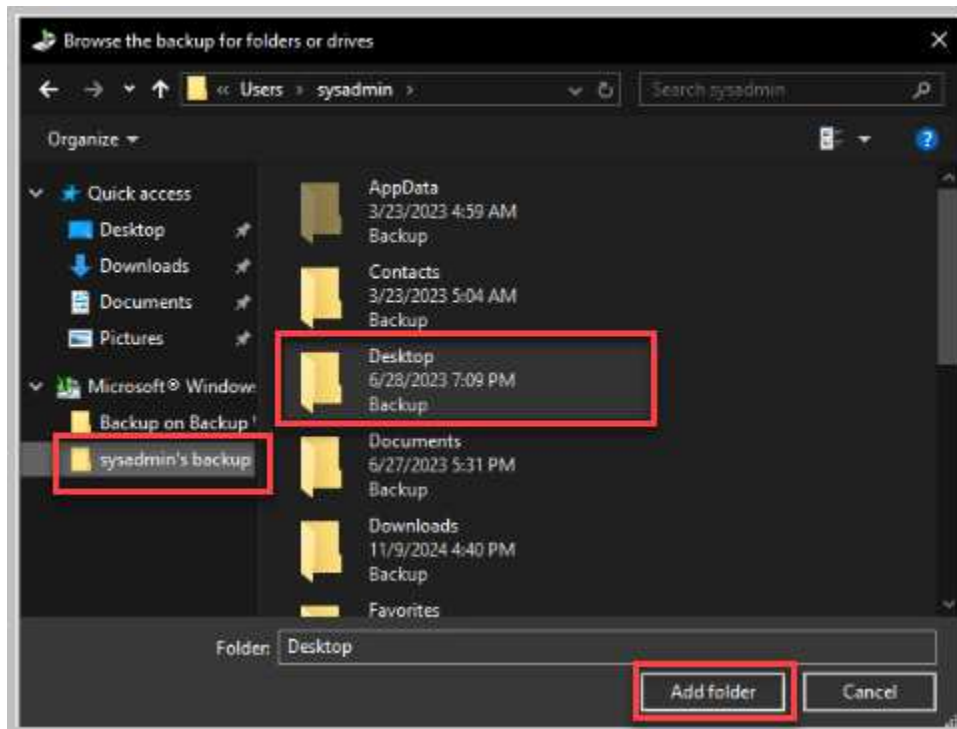
14. Change focus to the **Backup and Restore** window. Click on the **Restore my files** button.



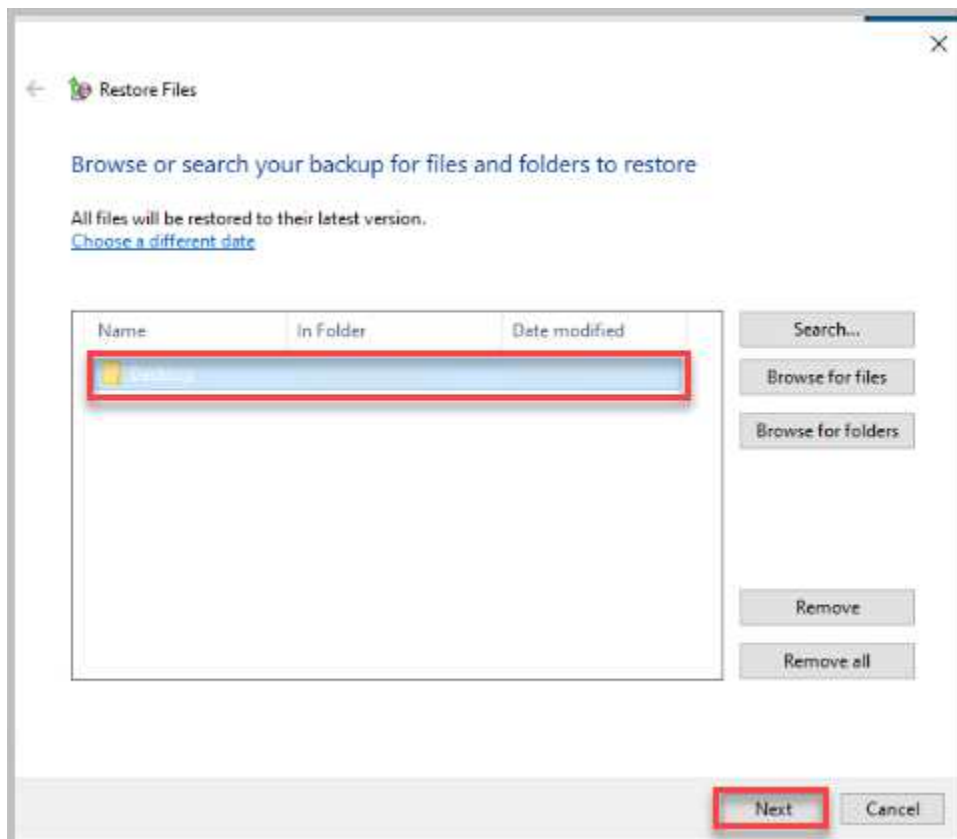
15. In the *Restore Files* window, click on the **Browse for folders** button.



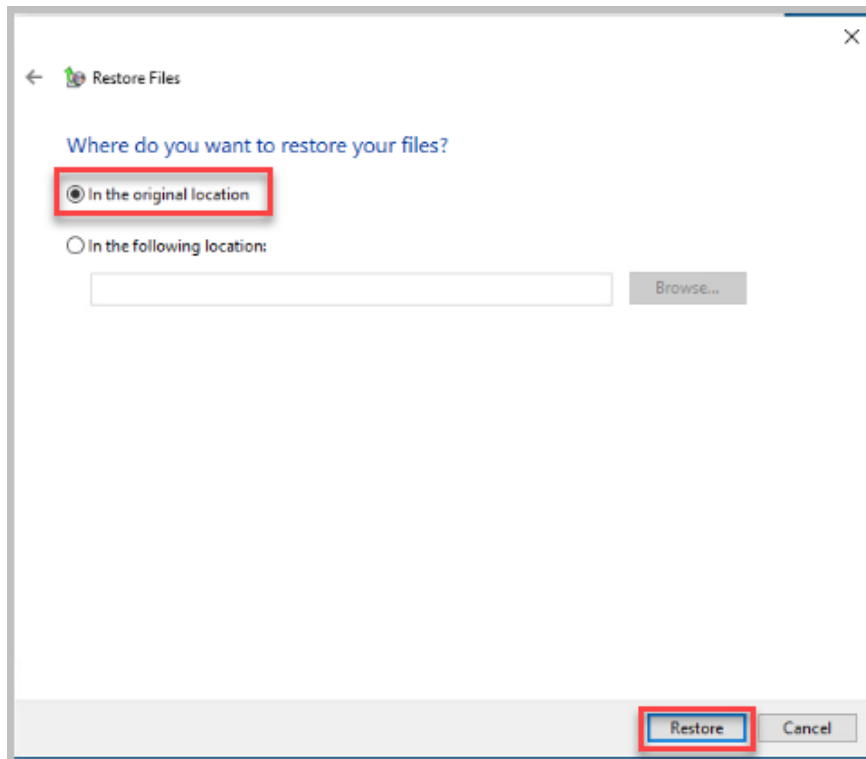
16. On the left pane, click on the **sysadmin's backup**, followed by clicking on the **Desktop** folder in the right pane. Select **Add folder**.



17. Back on the *Restore Files* window, notice the *Desktop* folder is listed. Click **Next**.



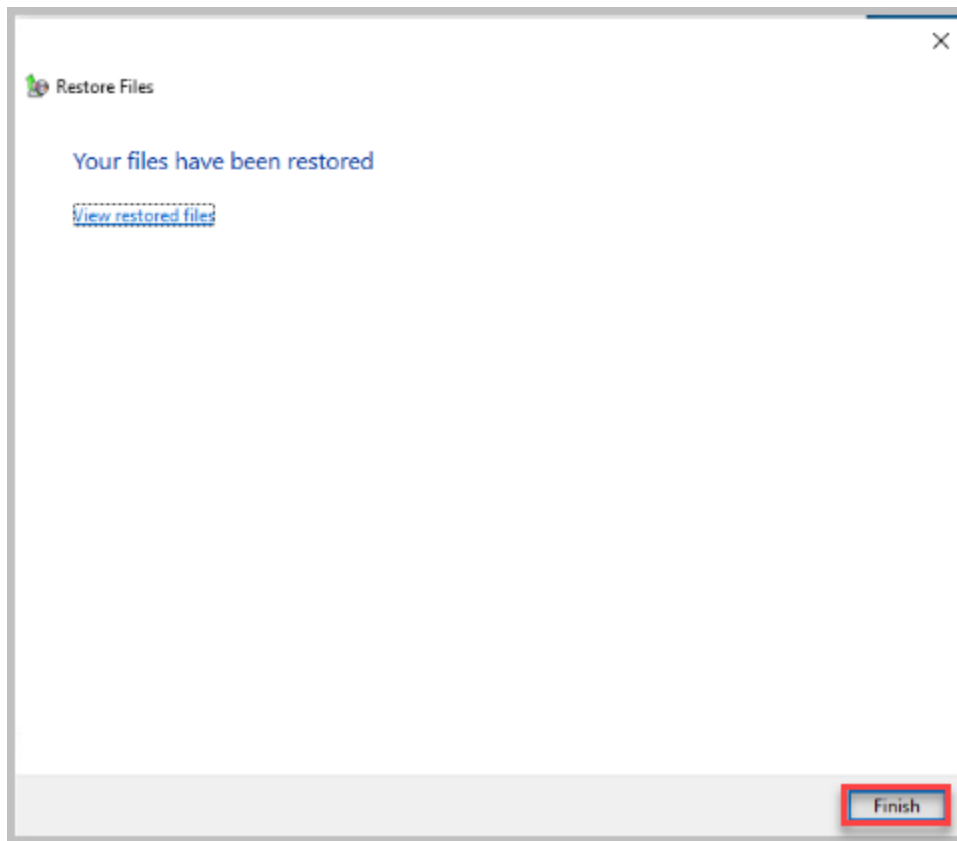
18. When asked where to restore the files, ensure that **In the original location** radio button is selected and click the **Restore** button.



19. If prompted, Select **Do this for all conflicts**, and select **Copy and Replace**.



20. Once the *Restore Files* process completes, click **Finish**.



21. Notice the *Toolbox shortcut* is back on the *Desktop*. Close *Backup and Restore*. Leave the Win10-OS system open for the next activity.



Challenge Task:

As a challenge, try to recreate this task in the *Win11-OS* machine that is made available in this lab environment. This will help show similarities and differences between the versions.

4.3 Initiating a Backup Process from Win10-OS to Win11-OS

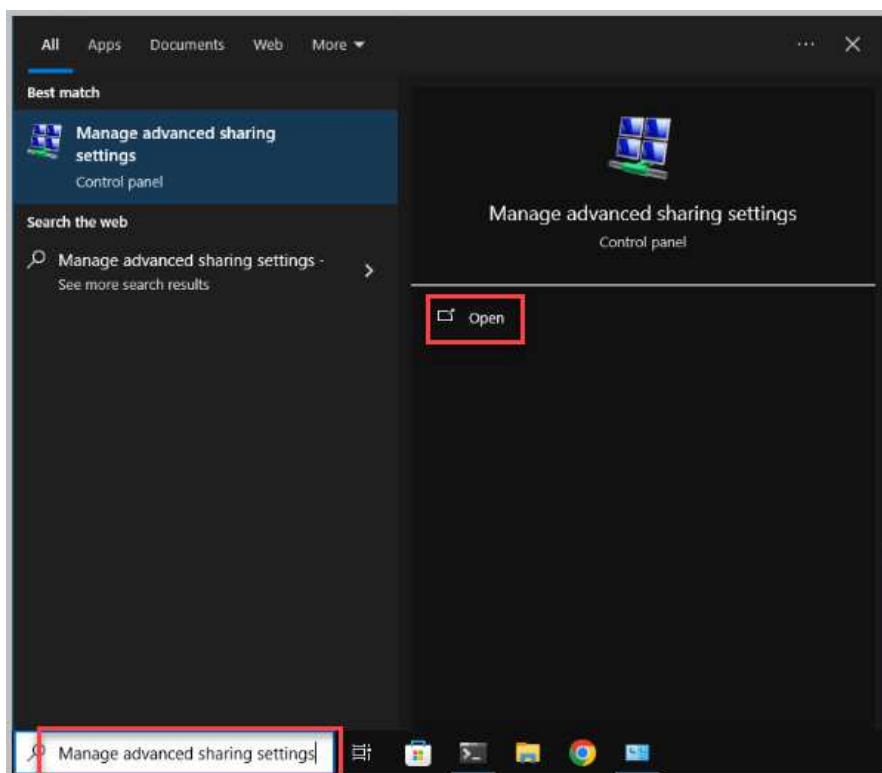
Windows 10's File History is an easy way to get started backing up your personal files, since it's built into your system. It allows you to recover files that might have been accidentally deleted or to restore a modified document to an earlier version.

Backing up data prevents you from losing it. This can be done by using a USB flash drive or a valid network-shared or network-attached drive.

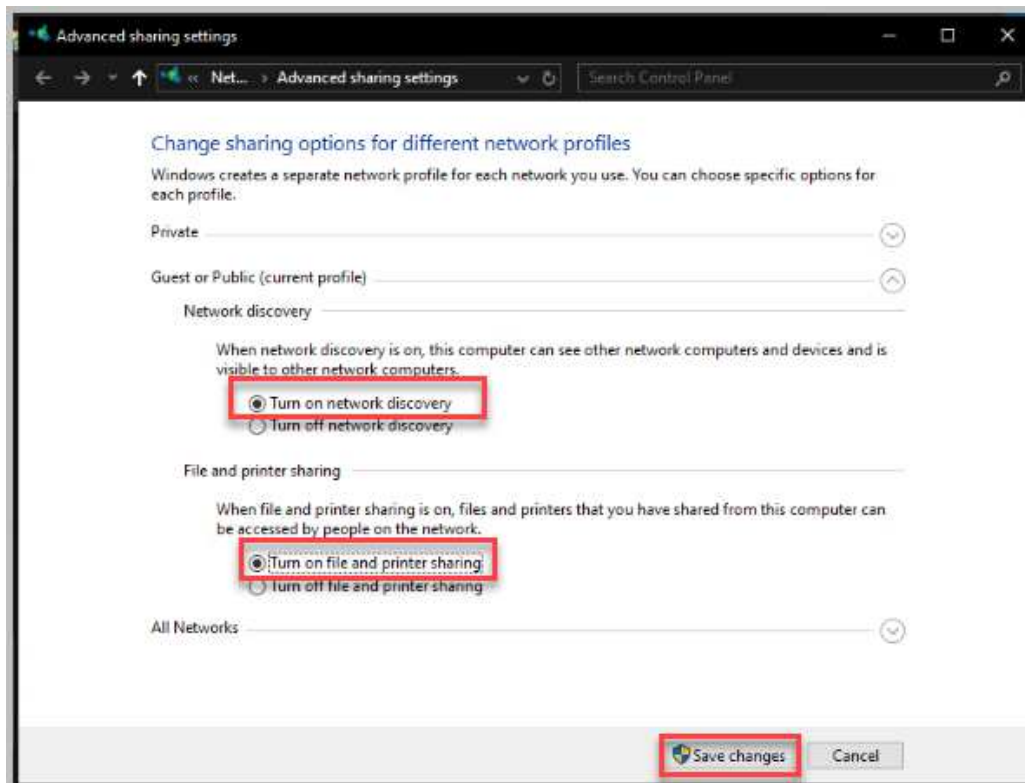
Windows 10's File History is a good option for a PC backup strategy, but it is only one part. It is safest to store data in three places: the working copy on your internal hard drive, a local backup, and a remote backup. The remote backup should be off-site, so if there is a local disaster, that copy remains safe in the remote location. The easiest way to handle remote backups is to use an online backup service. A backup is essentially just a copy of the data stored on your local drives or those on a network. Cloud storage is a remote storage option where you can store files outside of the local network, often on someone else's hard drives, that you can access over the internet.

In this exercise, you will use *Windows 10 File History* to back up to a network resource because cloud storage requires Internet access. Regardless of the tool used, the concept of backup is the same and necessary.

1. On the Win10-OS system, in the search bar on the taskbar, type **Manage advanced sharing settings**, and select **Open**.



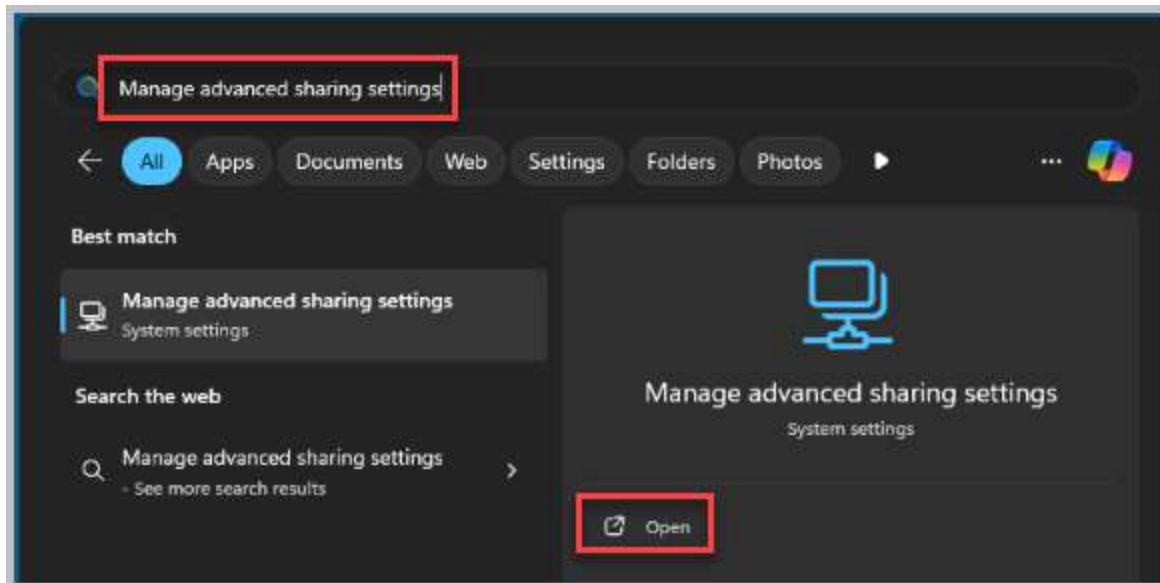
2. In Network Discovery under the *Private and the Guest or Public (current profile)*, select **Turn on network discovery** and select **Turn on file and printer sharing**. Click **Save Changes**.



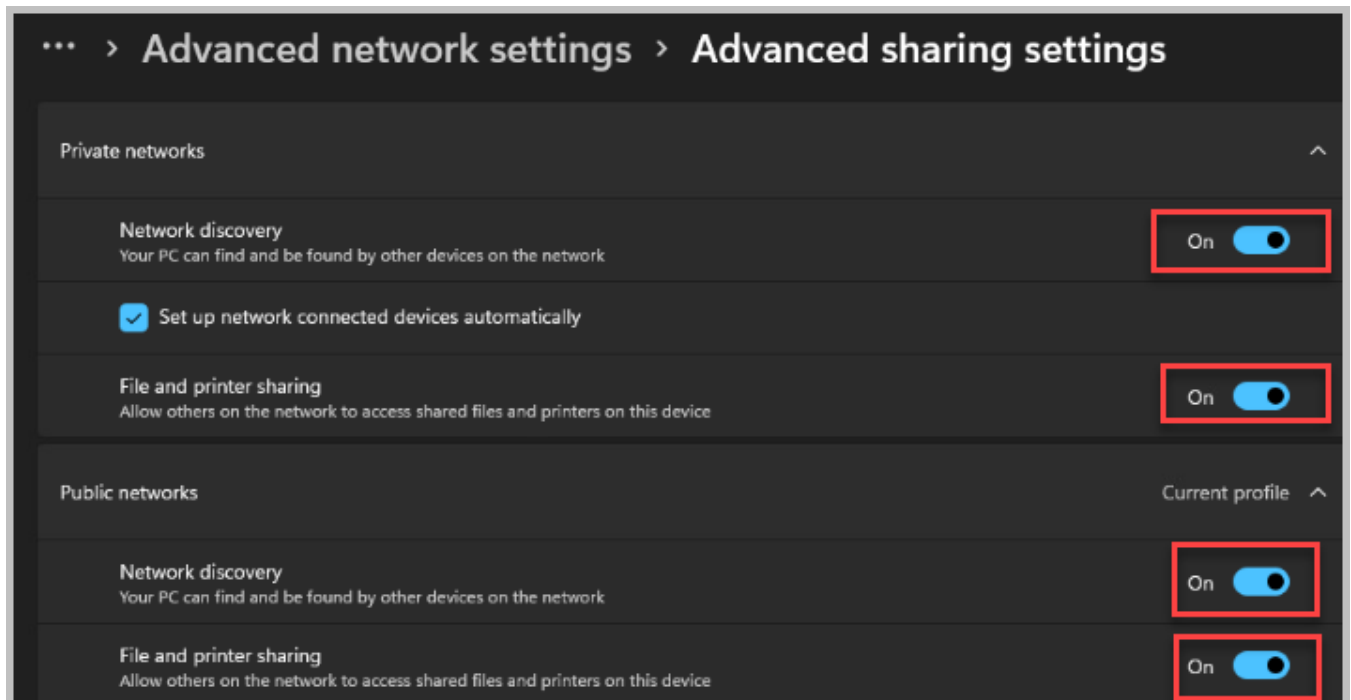
3. Switch to the **Win11-OS** console.



4. Select **Search** and type **Manage advanced sharing settings**. Select **Open**.



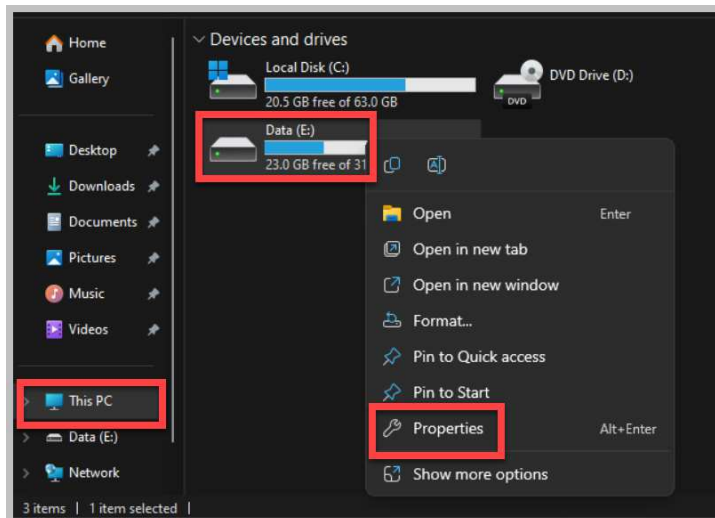
5. In Advanced sharing setting under the *Private and the Public networks*, select **Turn on network discovery** and select **Turn on file and printer sharing**.



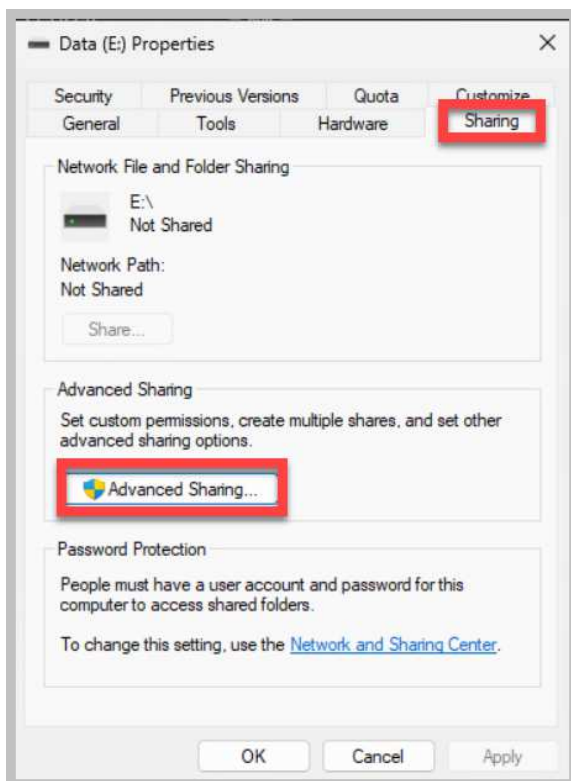
6. Close open windows.
7. Select **File Explorer** on the taskbar.



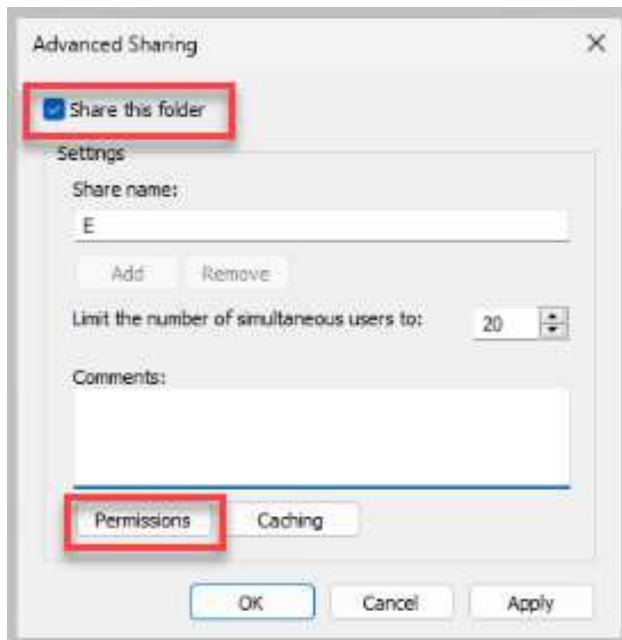
8. Select **This PC** and right-click on the **Backup (E:) drive**. Select **Properties**.



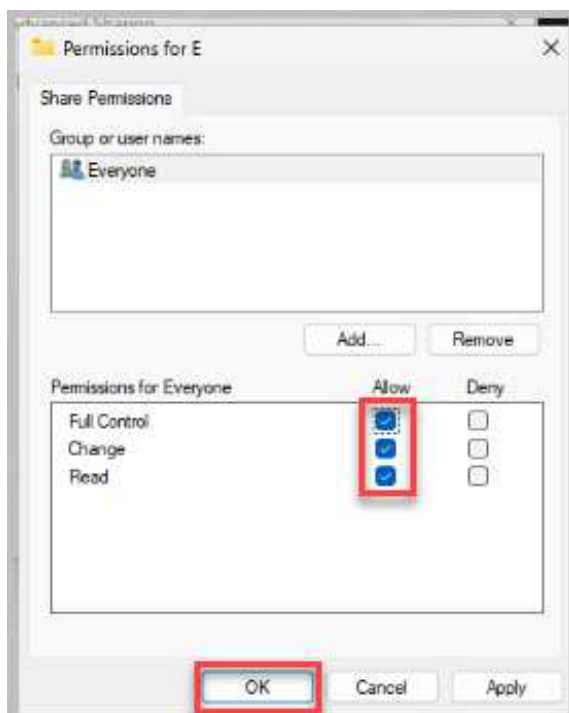
9. Select the **Sharing Tab**, and select **Advanced Sharing**.



10. Select **Share this folder** and select **Permissions**.



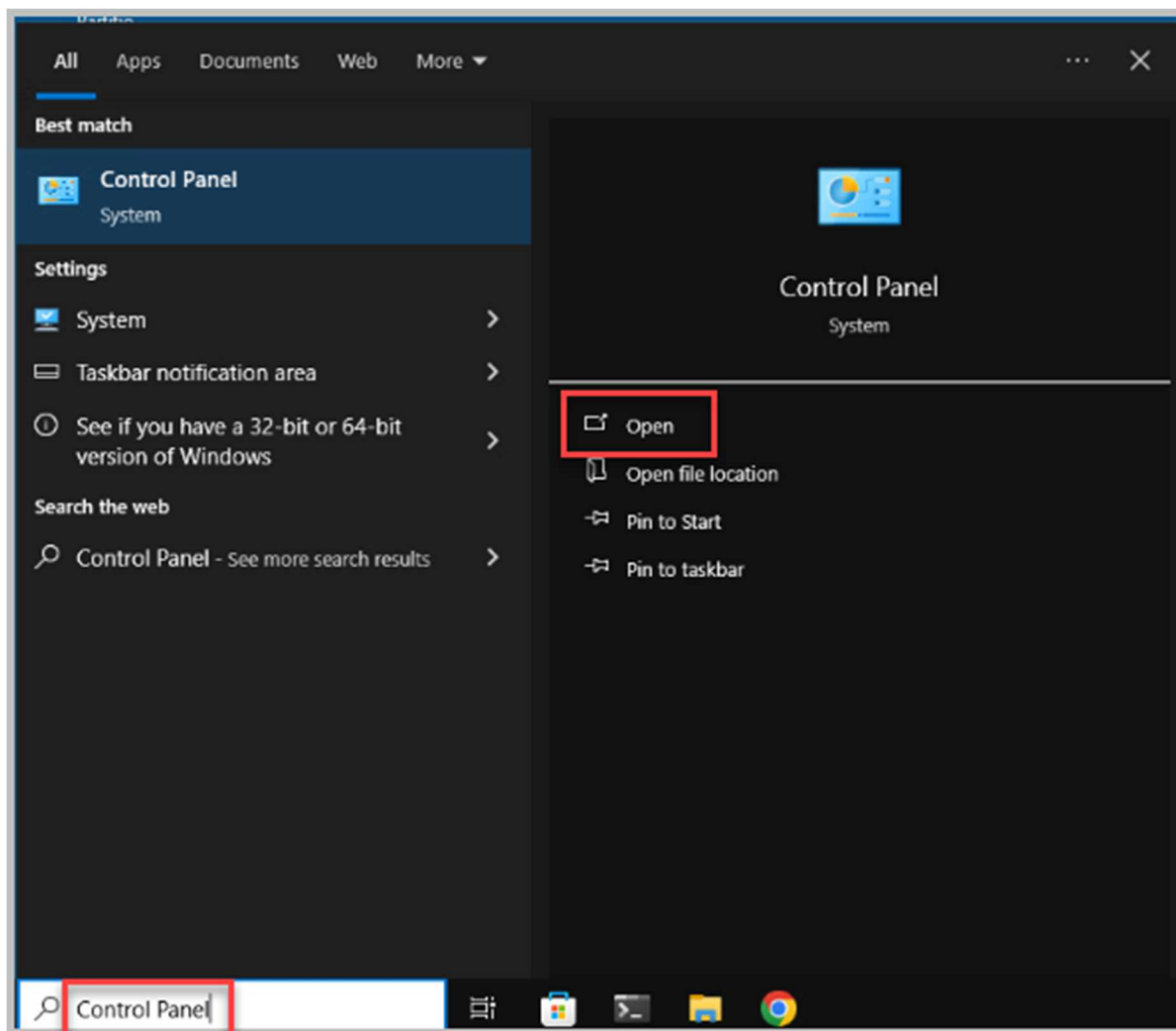
11. Change *Permissions for Everyone* by selecting **Full Control**. Select **OK** two times. Close any open windows.



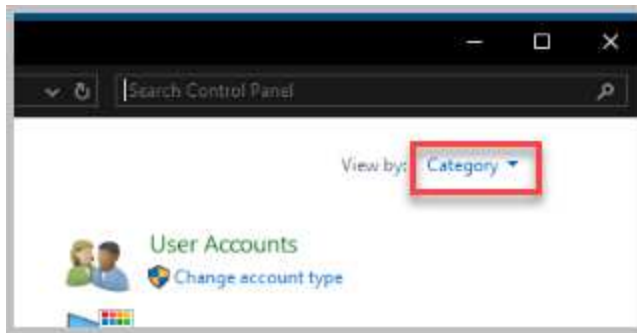
12. Switch to the **Win10-OS** Console connection.



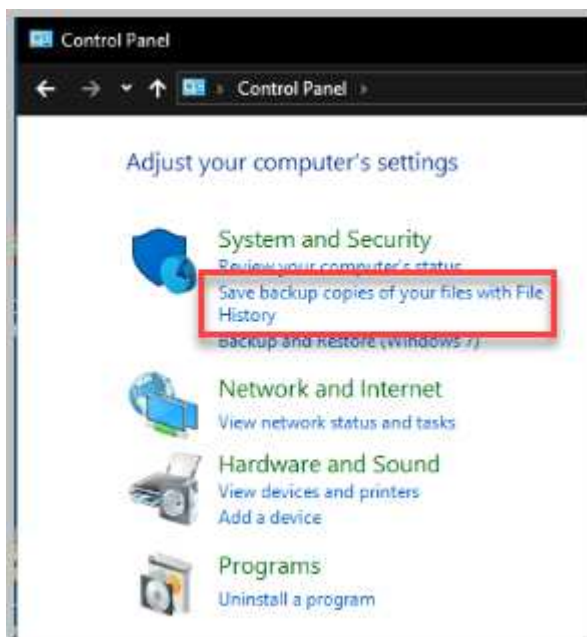
13. **Select** the search bar on the taskbar, type the command **Control Panel**, and select **Open**.



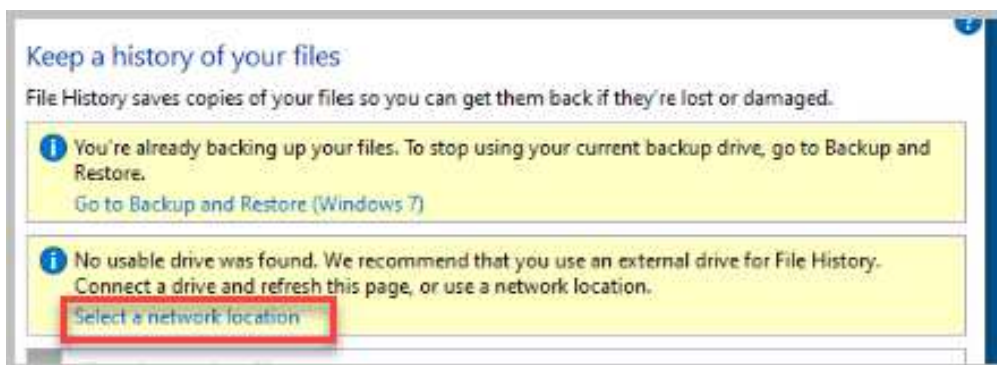
14. In the *Control Panel* window, change the *View* to **Category**.



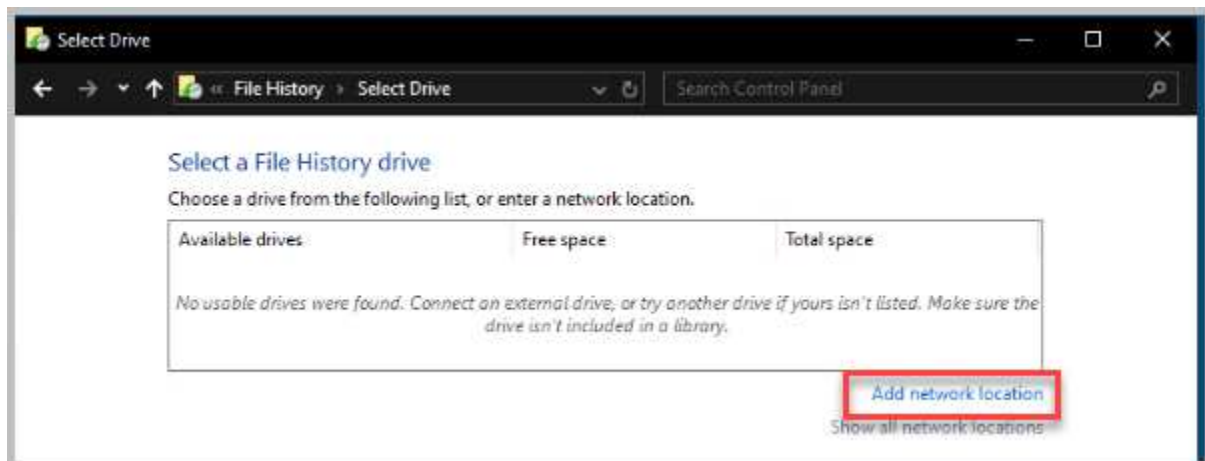
15. Click on **Save backup copies of your files with File History**.



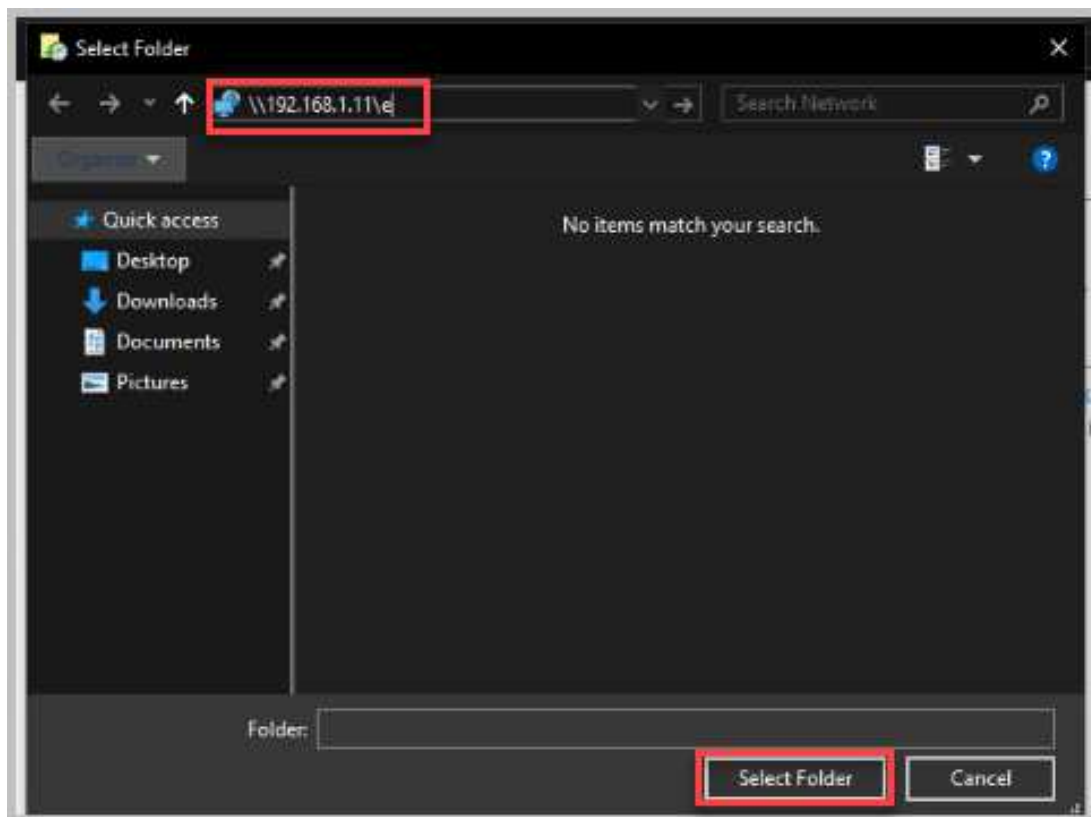
16. If there is no network location showing, click on **Select a network location**.



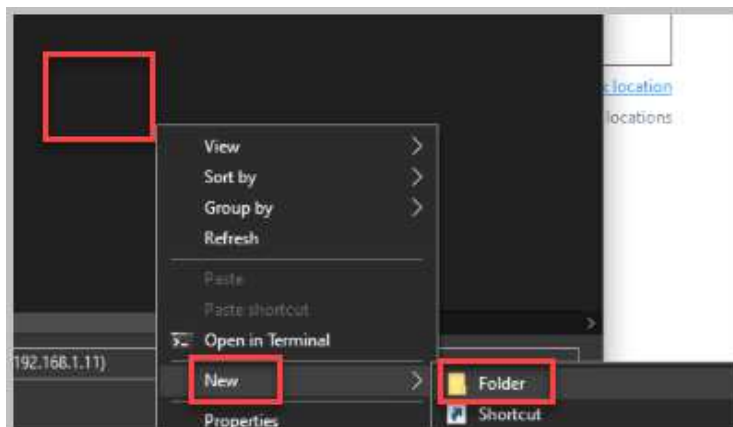
17. Click on **Add network location**.



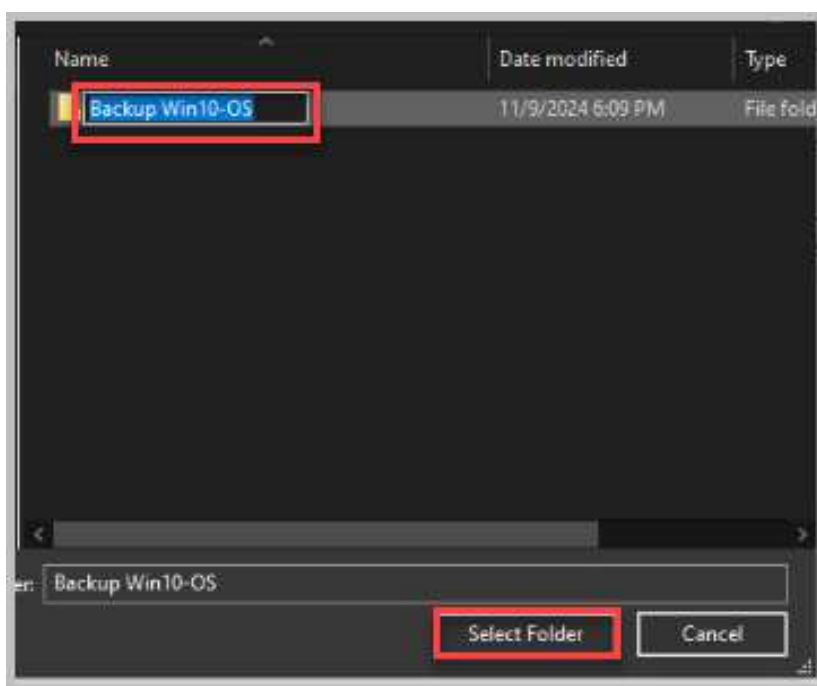
18. In the address bar, type `\\192.168.1.11\c` and click **Enter** on your keyboard.



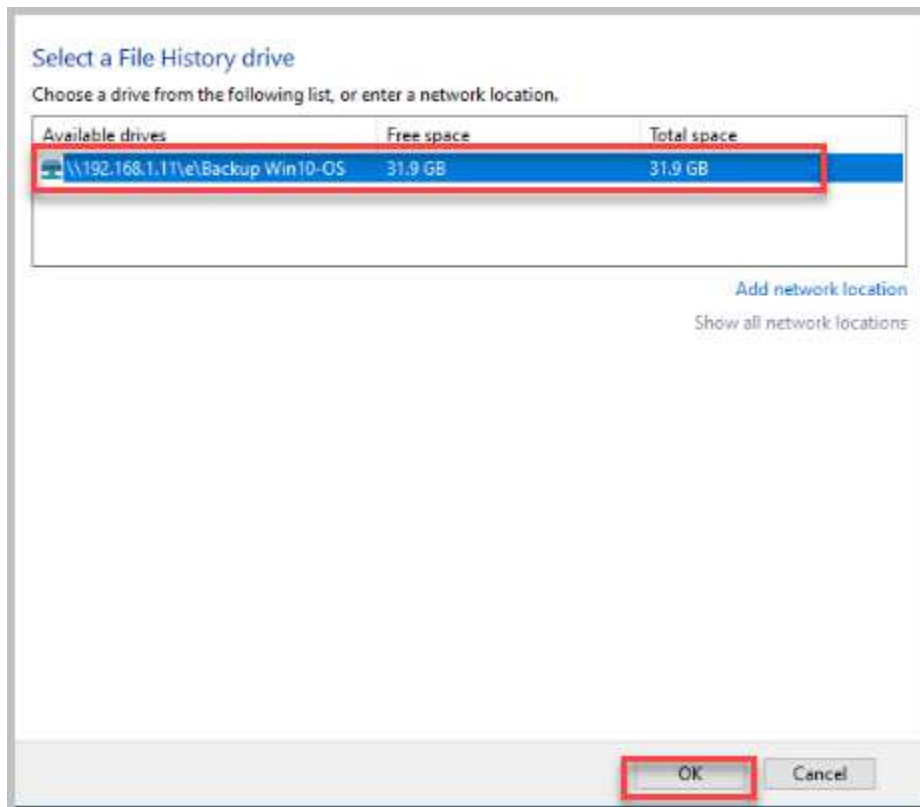
19. In the blank space of the *Network Shared Folder*, **right-click** and select **New**, select **Folder**.



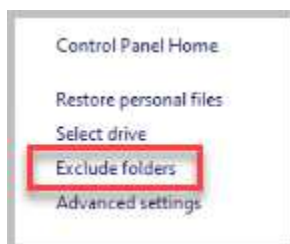
20. Type **Backup Win10-OS** and then click on **Select folder**.



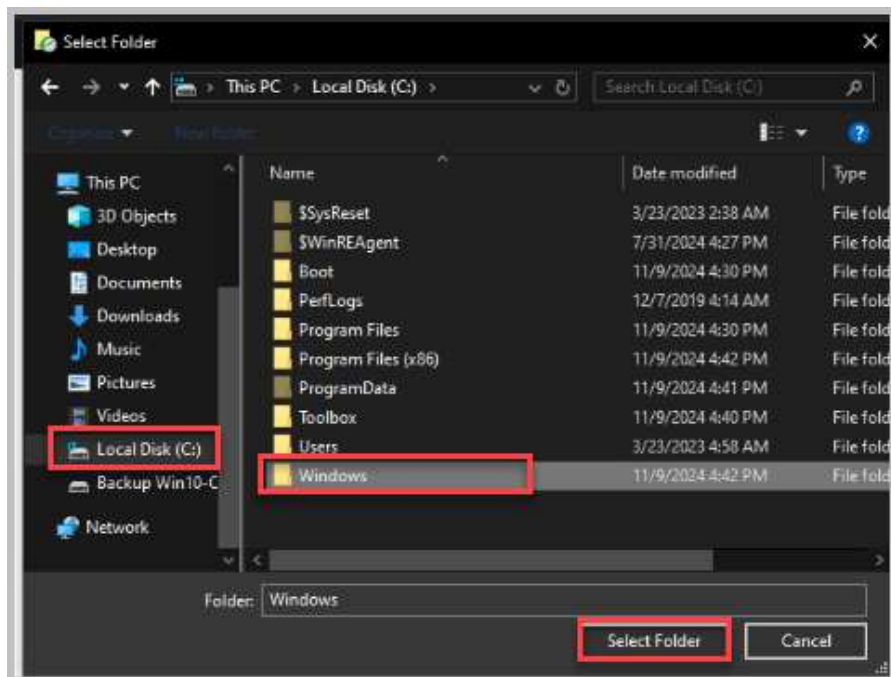
21. Click `\\192.168.1.11\e\Backup Win10-OS` to highlight it and click **OK**.



22. Click on **Exclude folders** from the left pane.



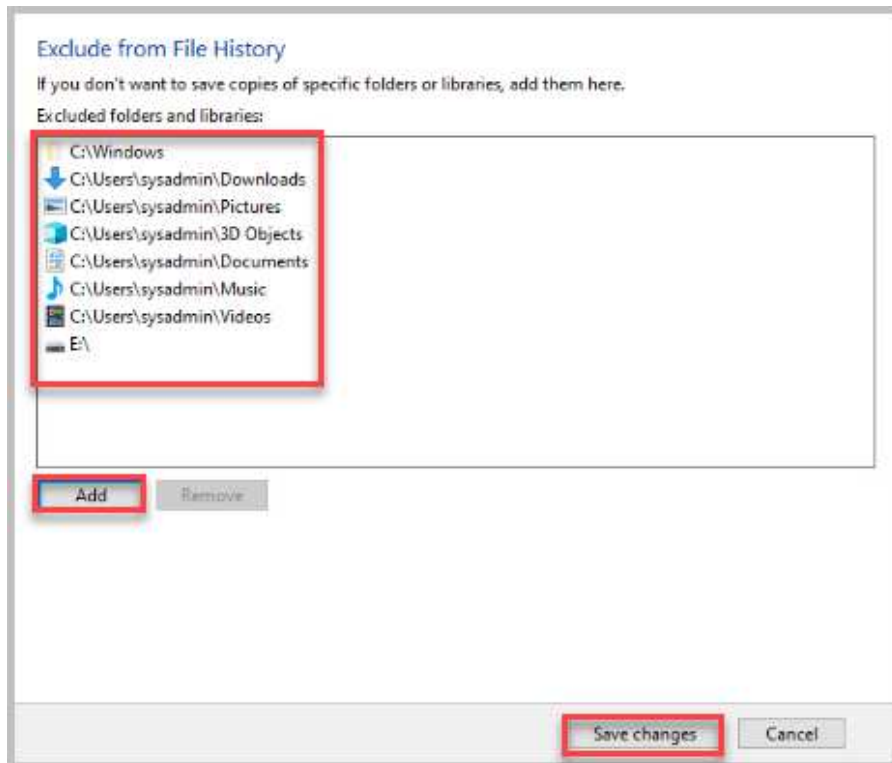
23. Click on **Add**. Select **Local Disk (C:)** and **Windows**. Click **Select Folder**.



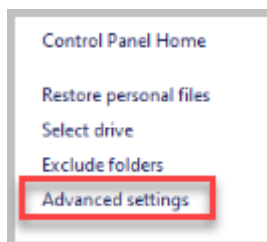
24. Select **Add**.



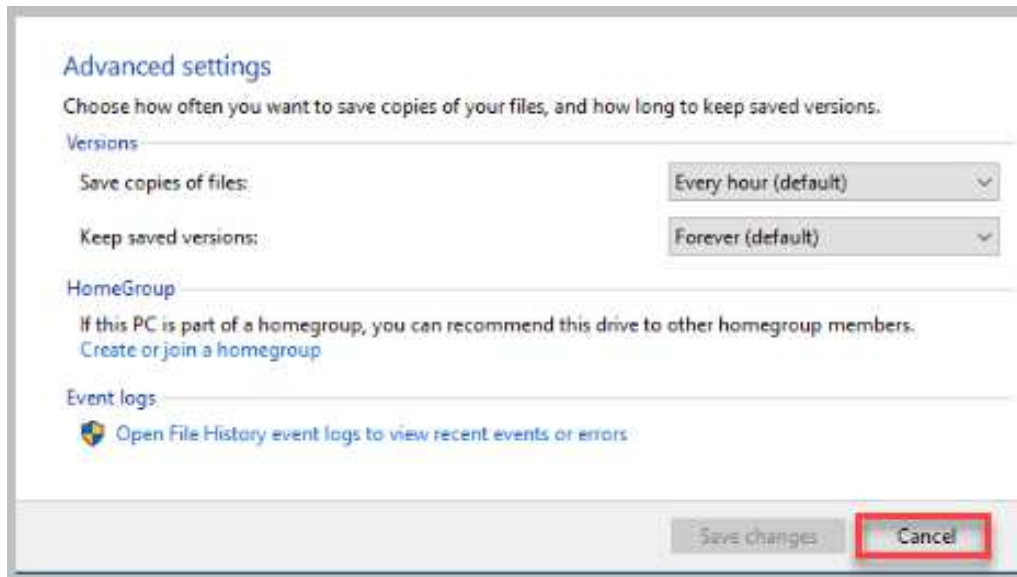
25. Click on each **Library** and **Add** all but the *Documents* Library. You will have to repeat this step for each library separately. Click **Save Changes** once all your selections have been added.



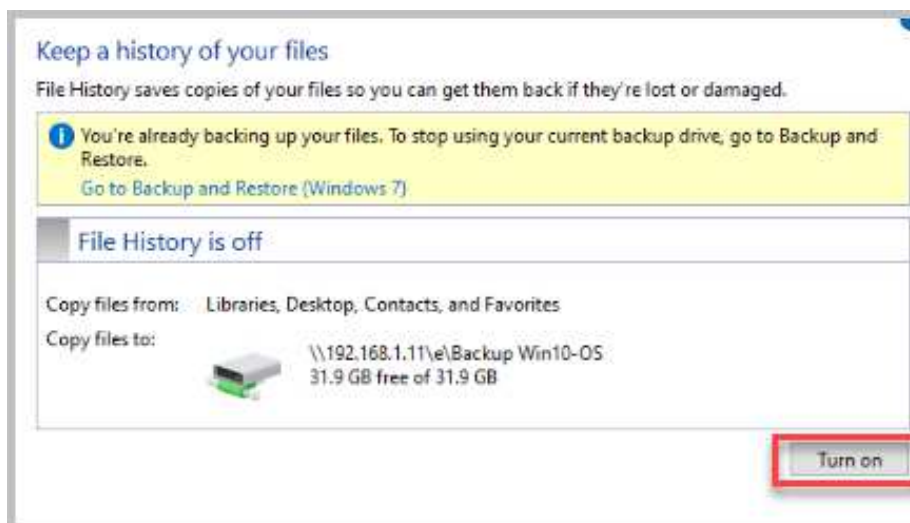
26. Click on **Advanced settings** from the left pane.



27. Click through the options to see how you could personalize the backup options. Once finished reviewing, click **Cancel**.

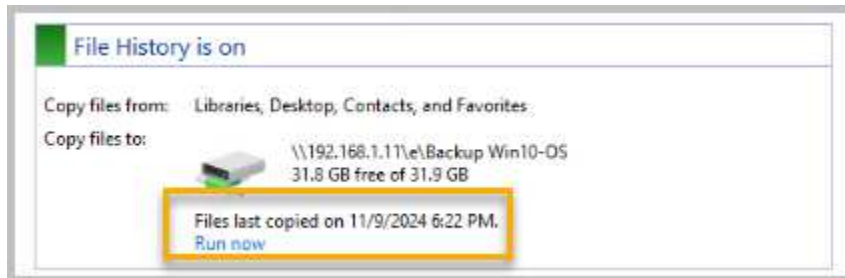


28. Click the **Turn on** button to enable *File History* and to start the backup process.



The information note in the yellow strip above may show up when you are backing up files that have been encrypted with EFS. File History will not back up sensitive data.

29. It will take a few minutes to complete the backup, but you will see the timestamp of when the backup is completed. *Run now* will become available to do manual backups. Click the **X** in the top-right corner to close.

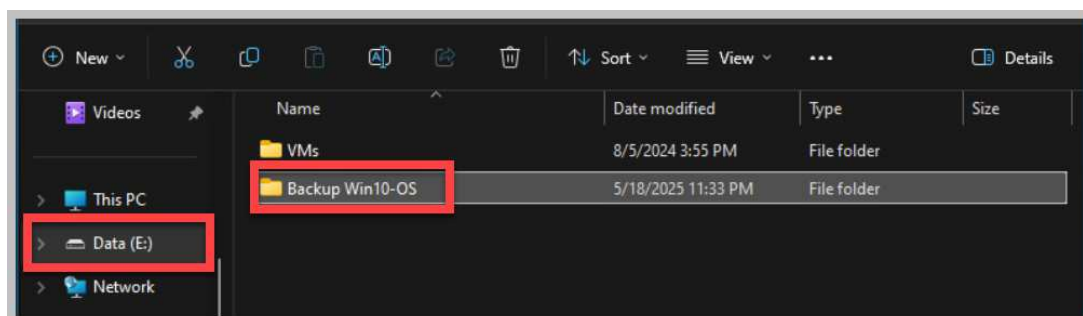


4.4 Log in to the Win11-OS System to Find the Backup.

1. Select **File Explorer** on the Taskbar.



2. Select **Data (E:)** and double click the **Backup Win10-OS** folder.



3. Notice the **Sysadmin** folder has been saved correctly.



Notice the **sysadmin** folder is present. If you explore the Data directory further, you'll notice that the backup files are located here.

4. The lab is now complete; you may end the reservation.